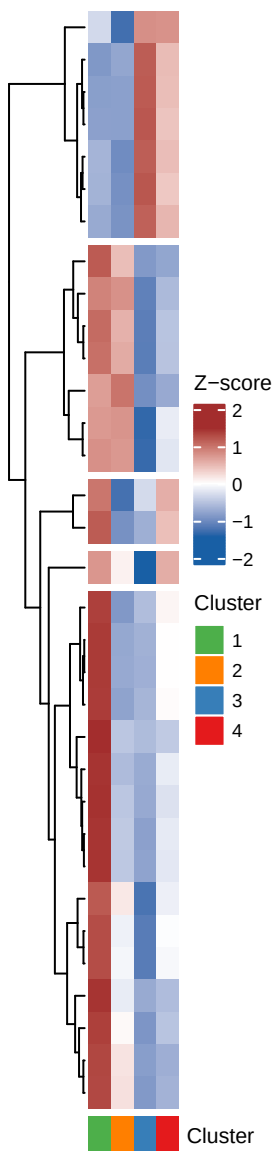
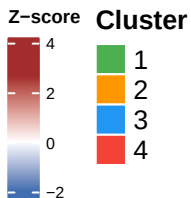
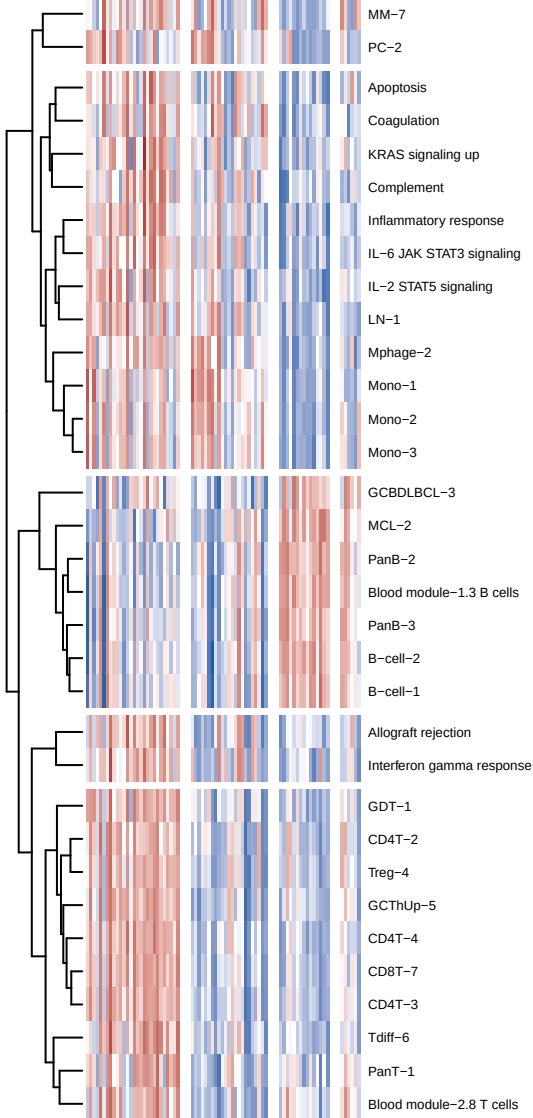


GSVA scores (RNA)

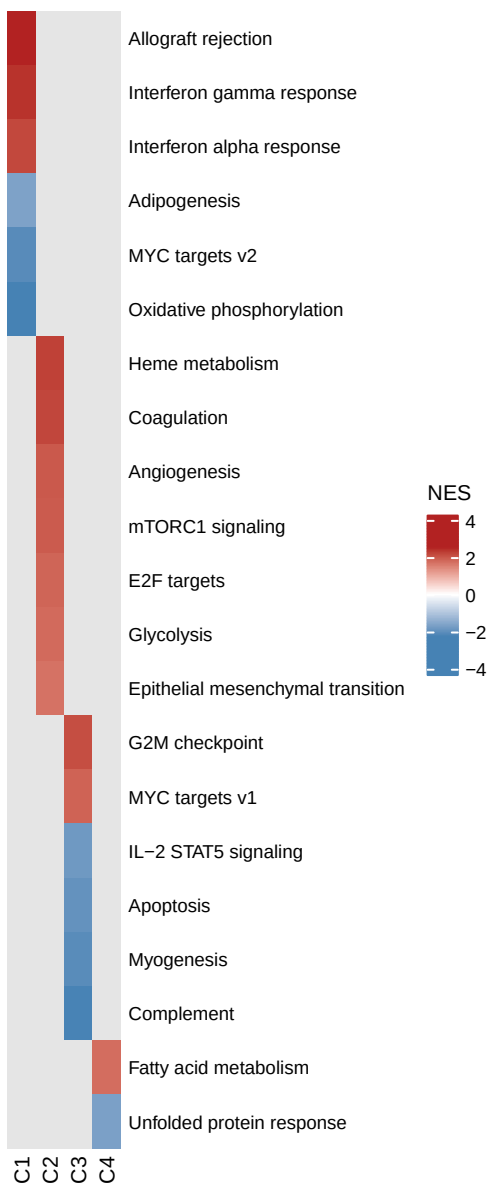


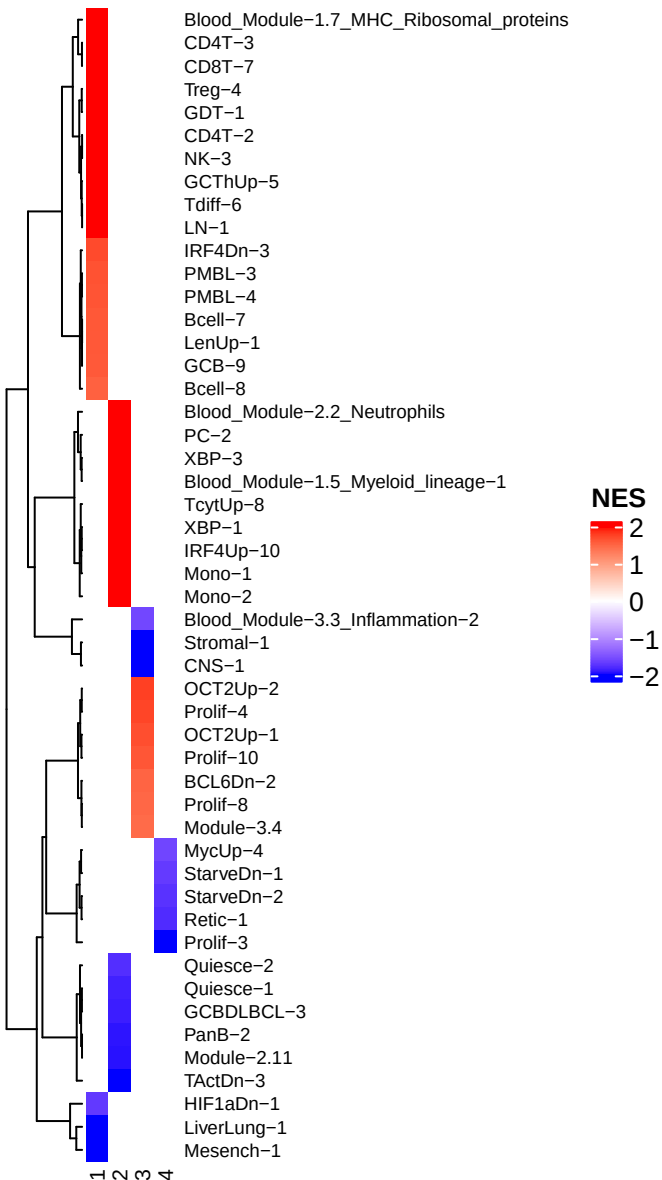
1 2 3 4

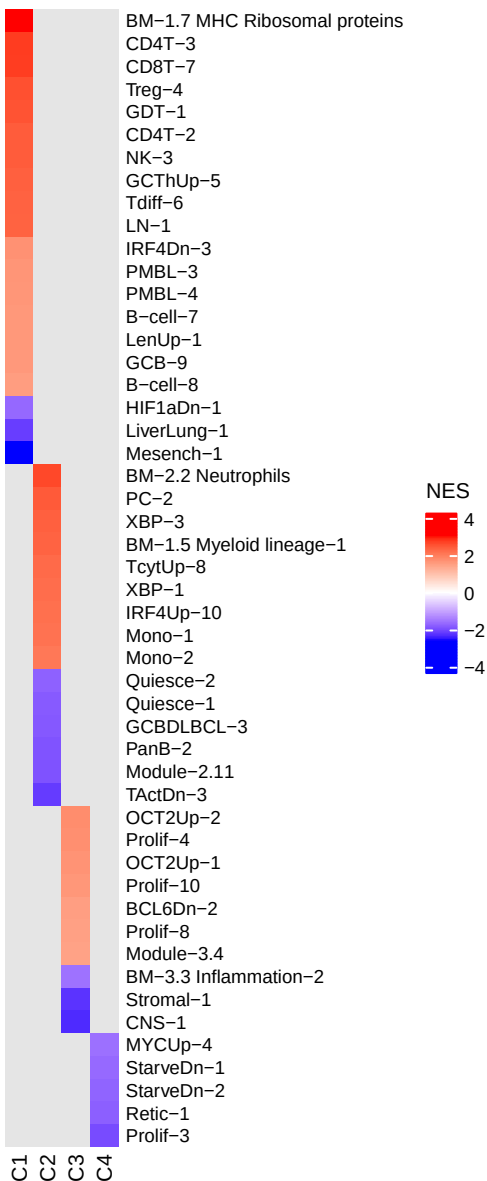
Cluster

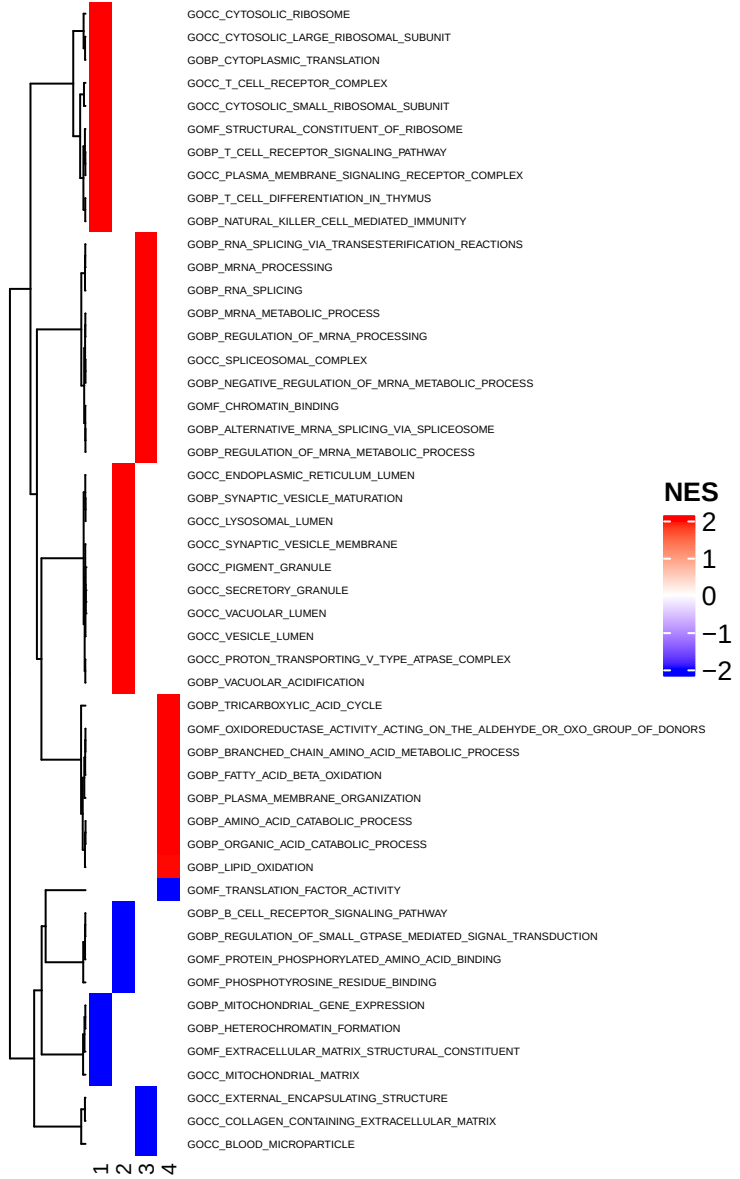


Expression by Cola Classes (k=4, Z-scored)																							
935_07	935_02	935_07	935_10	930_06	930_08	753_07	757_01	772_07	772_06	775_06	775_05	764_01	753_08	757_04	772_04	757_06	753_09	757_09	772_01	764_10	772_01	764_10	772_01
-0.7	-0.4	-1.1	-1.0	2.2	0.9	-0.5	-0.9	-0.1	0.8	0.9	-1.0	0.7	0.0	-1.9	0.1	-0.8	-0.4	0.1	-0.5	-2.0	-0.5	-2.0	-0.5
-2.0	1.5	-1.0	-1.3	0.6	0.5	0.2	0.6	0.8	-0.4	-2.0	1.2	-0.0	1.0	0.5	-0.6	0.2	0.6	0.0	-1.2	-2.0	-1.2	-2.0	-1.2
-0.2	0.4	0.0	-2.9	-2.9	-0.2	-1.3	0.5	1.5	-0.5	0.6	0.3	0.6	1.1	-1.7	0.8	0.2	0.2	0.5	0.3	-1.0	0.3	-1.0	-1.0
-1.1	-0.8	-1.4	-0.7	0.2	-0.4	1.0	0.3	0.9	0.7	-0.2	-1.7	0.3	0.1	0.5	1.0	0.2	0.4	0.4	1.1	-0.0	1.1	-0.0	-0.0
0.1	0.1	-0.2	-0.3	0.2	-0.6	1.4	-0.2	1.3	0.4	0.6	0.4	0.3	0.5	-0.6	-0.5	0.4	0.4	-0.8	-1.0	-2.0	-1.0	-2.0	-1.0
0.6	0.2	-1.7	-2.3	-0.8	-0.3	-0.0	0.7	-0.2	1.4	0.9	0.1	-1.0	1.0	-1.2	-0.6	-0.6	-0.4	-3.0	-2.0	-0.0	-2.0	-0.0	-0.0
0.8	0.6	-1.7	-2.6	-1.0	-1.3	1.3	0.9	0.6	0.9	1.7	-0.9	-2.6	-0.2	0.3	-1.4	0.1	0.0	-0.9	-0.5	1.1	-0.5	1.1	-0.5
-0.0	-0.6	-1.8	-2.8	-0.7	-0.5	1.4	-0.3	0.3	-0.1	2.8	0.7	-0.0	0.7	1.6	-1.8	2.3	1.1	-1.1	-0.2	-0.0	-0.2	-0.0	-0.0
0.1	0.8	0.0	-0.4	2.0	-0.6	1.1	-0.7	-1.6	-0.9	0.4	0.4	-0.7	0.0	0.2	1.1	-0.5	-1.8	0.1	1.1	1.5	1.1	1.5	1.5
1.0	1.3	0.6	0.8	1.1	-0.4	1.4	-0.2	-3.0	-1.3	-1.0	2.1	-0.9	1.1	0.7	1.1	0.4	-0.3	1.0	1.0	1.3	1.0	1.3	1.3
1.7	1.1	0.9	1.5	0.5	0.3	1.3	0.5	-0.5	-0.5	-0.8	0.5	-0.1	0.2	3.5	1.8	0.3	-0.7	-1.1	1.0	-0.0	-0.0	-0.0	-0.0
1.5	1.1	1.4	1.9	0.3	-0.1	0.0	0.9	1.4	-1.1	0.1	0.4	0.4	0.0	0.4	0.2	-0.4	-1.4	1.1	0.8	-0.0	-0.0	-0.0	-0.0
1.4	1.0	0.2	0.8	-2.7	1.6	1.3	0.1	0.1	-0.6	-0.4	0.9	-0.5	1.8	2.8	-0.5	1.7	0.6	-0.2	-0.2	0.5	-0.2	0.5	0.5
-0.5	-0.4	-0.9	-0.8	1.4	-0.2	1.8	-0.1	-1.4	-0.7	-1.4	-0.7	1.9	0.6	-0.7	-0.2	1.7	1.4	-0.5	0.8	0.7	0.7	0.7	0.7
-0.2	0.7	0.0	0.8	0.2	0.0	1.6	-2.4	-0.2	0.7	0.2	-0.4	0.1	-0.1	0.2	-0.6	0.5	-0.3	0.2	-0.1	-1.0	-0.1	-1.0	-1.0
-0.2	0.2	-0.0	0.9	-0.9	1.1	1.7	-0.6	-1.1	0.2	-0.0	0.9	0.0	0.4	0.4	-0.3	1.3	0.7	0.4	1.1	0.2	0.2	0.2	0.2
-1.2	-1.4	0.8	1.3	-0.5	0.6	1.1	-1.1	-0.8	-1.0	0.5	0.9	1.5	0.1	0.4	-0.3	0.1	0.5	0.3	-0.2	-2.0	-0.2	-2.0	-2.0
-0.1	0.3	1.5	1.6	-0.8	1.7	1.2	0.4	-2.1	1.5	0.6	0.3	1.4	0.4	0.8	-0.0	1.2	0.8	-0.2	-1.4	-1.0	-1.0	-1.0	-1.0
-0.9	-0.5	0.4	-1.3	-1.3	1.1	0.1	0.1	-0.0	-1.4	-0.2	-2.1	0.4	0.4	-0.5	1.1	-1.0	-0.6	0.3	-0.3	-0.0	-0.0	-0.0	-0.0
-0.1	0.5	0.4	-0.9	-1.6	1.3	-0.5	0.1	-0.5	0.6	0.1	-0.6	1.8	0.0	-1.6	0.1	-0.7	0.7	-0.5	-0.5	-1.0	-1.0	-1.0	-1.0
0.7	0.7	1.1	0.9	-2.4	0.4	-0.8	-1.4	-0.6	-0.3	0.2	-0.5	1.0	0.4	0.8	0.4	-0.2	0.2	0.3	-0.5	-2.0	-0.5	-2.0	-2.0
-0.5	-0.5	1.0	0.9	-1.8	1.8	1.4	0.3	-0.3	0.7	0.7	0.3	0.2	1.4	-0.1	0.2	1.4	0.3	0.8	-2.2	-2.0	-2.0	-2.0	-2.0
-1.7	-0.8	-0.4	-0.8	-0.4	-1.8	-0.2	-1.7	0.0	-0.4	-0.8	0.1	1.5	-0.3	-0.5	-0.2	0.3	0.9	0.5	-0.8	0.7	0.7	0.7	0.7
-1.0	-0.5	-0.1	-2.8	-1.7	0.0	0.5	0.8	0.8	0.2	-0.3	0.8	1.9	-0.2	-0.6	-0.0	0.8	0.1	-0.7	-2.0	0.6	0.6	0.6	0.6
-0.0	-0.3	-0.6	-1.4	-0.8	3.2	-0.5	-0.4	0.6	-0.5	0.8	1.0	0.8	0.1	2.1	0.0	1.0	0.6	-0.6	-2.7	-1.0	-1.0	-1.0	-1.0
-0.5	-0.0	-0.3	-0.5	0.2	1.1	0.3	0.6	1.4	0.2	-0.5	2.8	1.0	0.4	-0.6	-0.5	-0.1	-0.1	-0.3	-0.8	-0.0	-0.0	-0.0	-0.0
-0.9	0.2	0.3	-0.4	0.5	0.8	-2.7	0.7	0.5	0.6	-0.3	-0.9	-0.0	1.2	-1.1	0.8	-2.5	0.1	0.5	0.2	1.8	0.2	1.8	1.8
-0.8	0.1	0.6	-0.6	-0.8	-0.1	-1.8	0.7	1.5	1.3	0.9	-1.8	-0.3	0.2	-0.5	0.3	-1.7	0.1	0.8	-2.0	-0.0	-0.0	-0.0	-0.0
0.6	0.5	1.2	0.9	-1.3	-0.7	0.8	-0.3	-0.6	0.7	0.1	0.7	-0.8	2.2	2.8	-0.5	1.5	0.1	-0.6	-0.8	-0.0	-0.0	-0.0	-0.0
-0.1	0.4	-0.0	2.1	0.8	-0.4	0.5	0.4	0.9	-0.8	-0.7	3.0	-0.6	0.1	2.4	0.2	0.3	-1.3	0.6	-0.3	3.6	3.6	3.6	3.6
-0.3	-0.1	-0.0	1.3	-1.3	-0.5	-0.5	-0.0	-0.2	-0.7	-1.1	4.0	-1.0	-0.3	-0.8	1.8	0.3	-0.7	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0
0.3	0.1	0.0	1.3	-0.6	-1.1	-0.3	0.2	0.1	-0.5	-0.9	2.7	-1.1	-0.7	0.9	1.8	0.0	-1.5	-1.4	-0.4	0.2	0.2	0.2	0.2
0.2	0.2	0.2	1.4	-0.8	-0.9	-0.1	-0.1	-0.1	-0.2	-0.8	2.5	-1.1	-0.5	1.1	1.3	-0.0	-1.5	-1.1	-0.5	-0.0	-0.0	-0.0	-0.0
-0.1	-0.9	-0.9	0.1	-0.7	0.7	0.3	0.7	0.8	0.0	0.6	-0.2	-0.2	-0.1	0.3	0.5	1.7	-0.5	-0.3	-2.5	-1.0	-1.0	-1.0	-1.0
-0.7	1.0	0.8	-0.5	-0.6	-0.5	-0.9	1.1	0.8	0.2	0.7	1.3	-0.8	-1.2	-0.2	1.5	0.1	2.1	-0.3	-0.4	-0.0	-0.0	-0.0	-0.0
-0.4	0.5	-0.2	-0.2	-0.3	0.6	0.9	1.0	1.2	1.1	-0.6	0.5	-0.3	0.2	0.6	1.4	0.9	-1.5	1.0	-1.1	-0.0	-0.0	-0.0	-0.0
-0.4	-0.3	-1.3	0.3	1.1	-0.4	-0.5	-0.7	-0.2	1.2	-0.4	2.0	-2.0	2.2	0.6	-0.7	-0.9	-0.7	2.0	-0.8	-1.0	-1.0	-1.0	-1.0
0.6	-1.5	0.8	0.2	0.3	0.9	2.6	1.2	-1.4	1.8	-0.5	-0.8	-1.6	0.8	-0.5	0.1	0.9	-1.5	-1.2	-0.5	1.0	1.0	1.0	1.0
0.2	1.4	0.7	0.1	-0.7	0.4	-0.2	0.7	1.0	0.3	-0.5	-0.8	-0.4	0.7	-1.1	-0.2	-1.1	-0.3	-0.2	-2.5	1.7	1.7	1.7	1.7
-0.1	0.4	0.6	-0.2	-0.1	0.1	-2.8	0.5	0.4	0.8	1.0	-1.7	-0.4	0.9	-0.2	0.1	-1.8	0.9	0.6	-1.6	-0.0	-0.0	-0.0	-0.0
-1.1	0.8	0.5	-0.5	-0.1	0.3	-2.3	0.6	1.3	1.4	-0.7	-0.9	-1.1	1.7	-1.7	0.5	-1.5	0.9	-0.4	-0.8	2.3	2.3	2.3	2.3
0.4	1.3	0.1	-0.6	-0.6	0.6	-0.9	-1.3	0.8	0.5	0.4	-0.6	0.2	-0.8	-1.8	0.5	-0.1	0.6	0.9	-0.4	-0.0	-0.0	-0.0	-0.0
-0.2	0.9	-0.7	-0.4	0.6	1.2	-1.9	-0.1	0.2	-0.6	-0.2	-1.6	-0.3	-1.0	-0.8	-0.5	-2.1	2.2	0.5	0.2	-1.0	-1.0	-1.0	-1.0







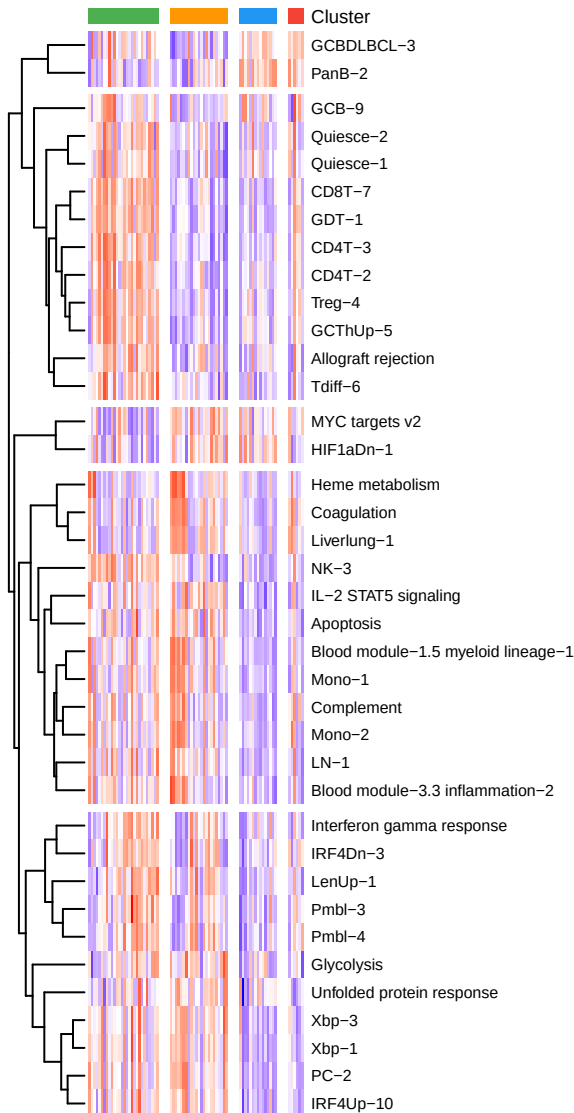


GSVA scores (Protein)

1 2 3 4

Cluster

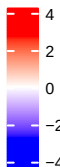
Protein



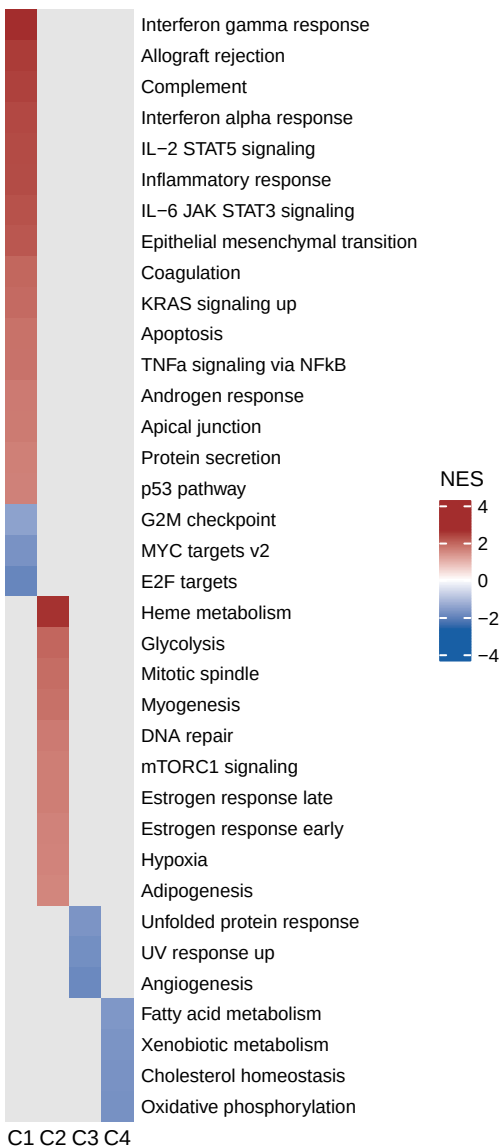
Cluster

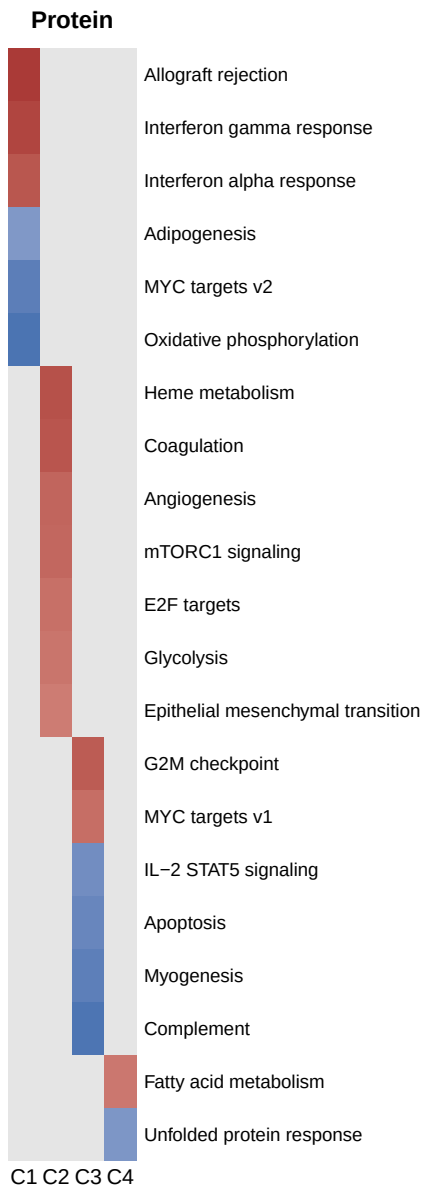


Z-score

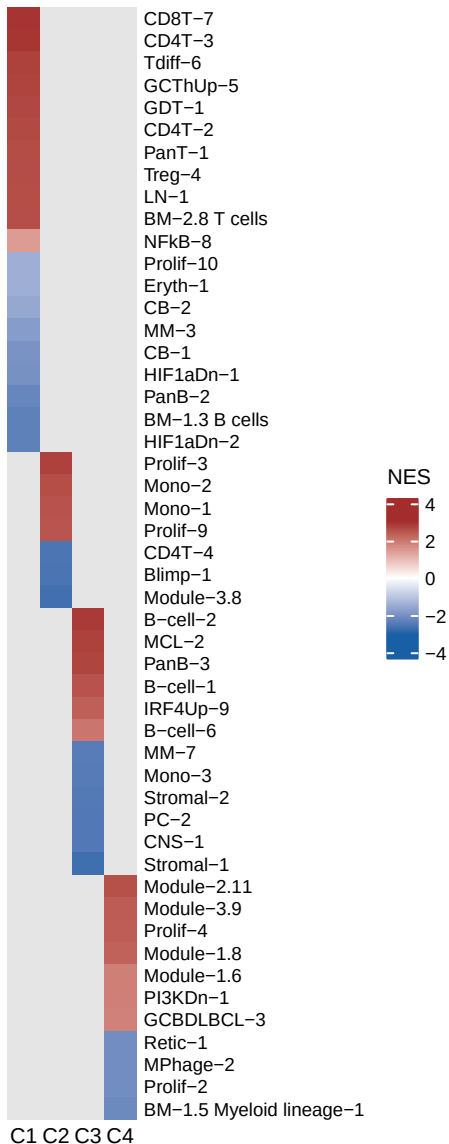


RNA

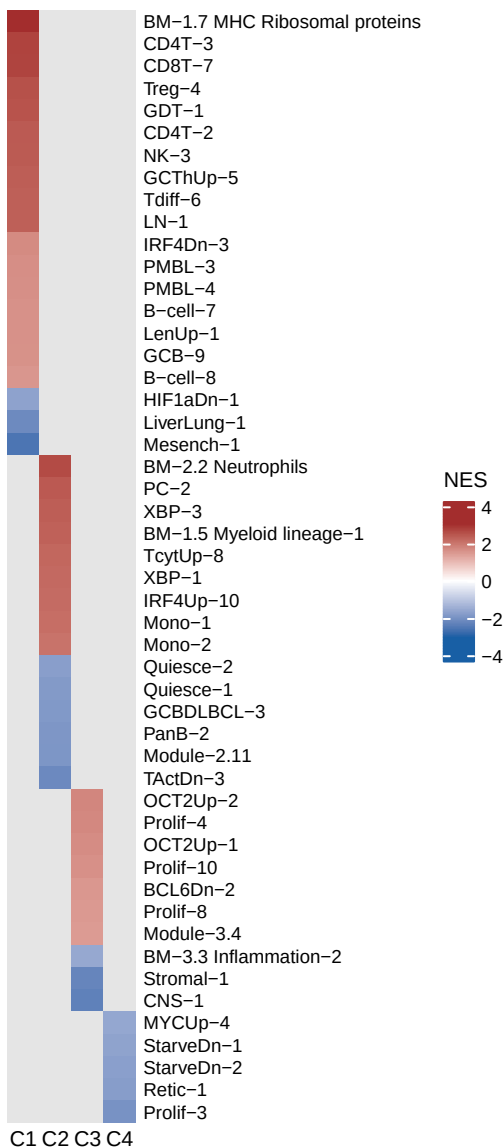




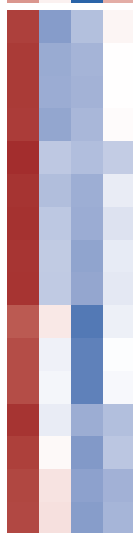
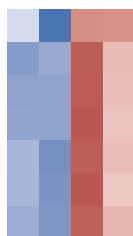
RNA



Protein

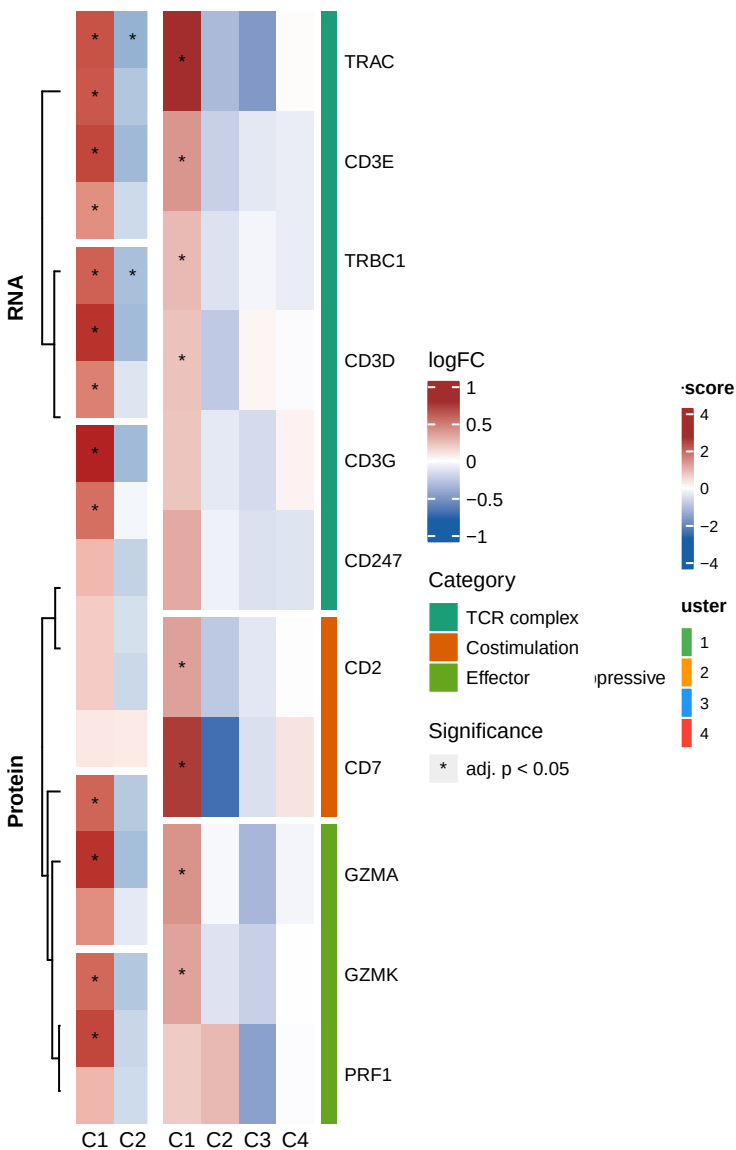


RNA

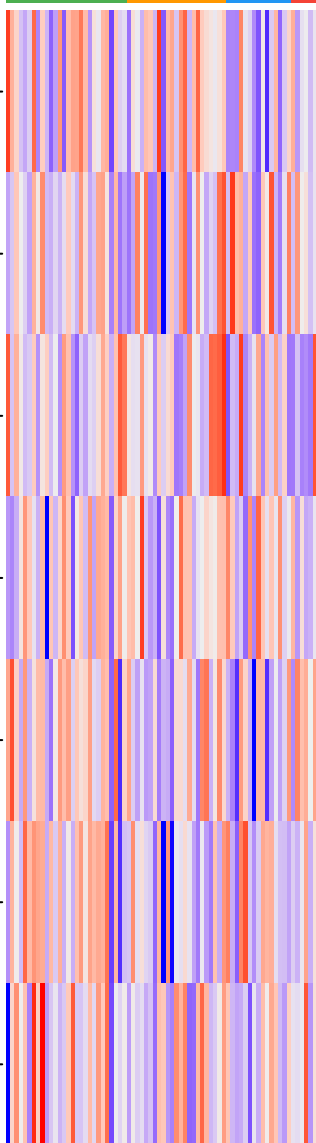


C1 C2 C3 C4

T cell T cell markers (Protein)



Cluster



AKT1

AKT2

RPS6KB1

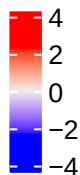
RPTOR

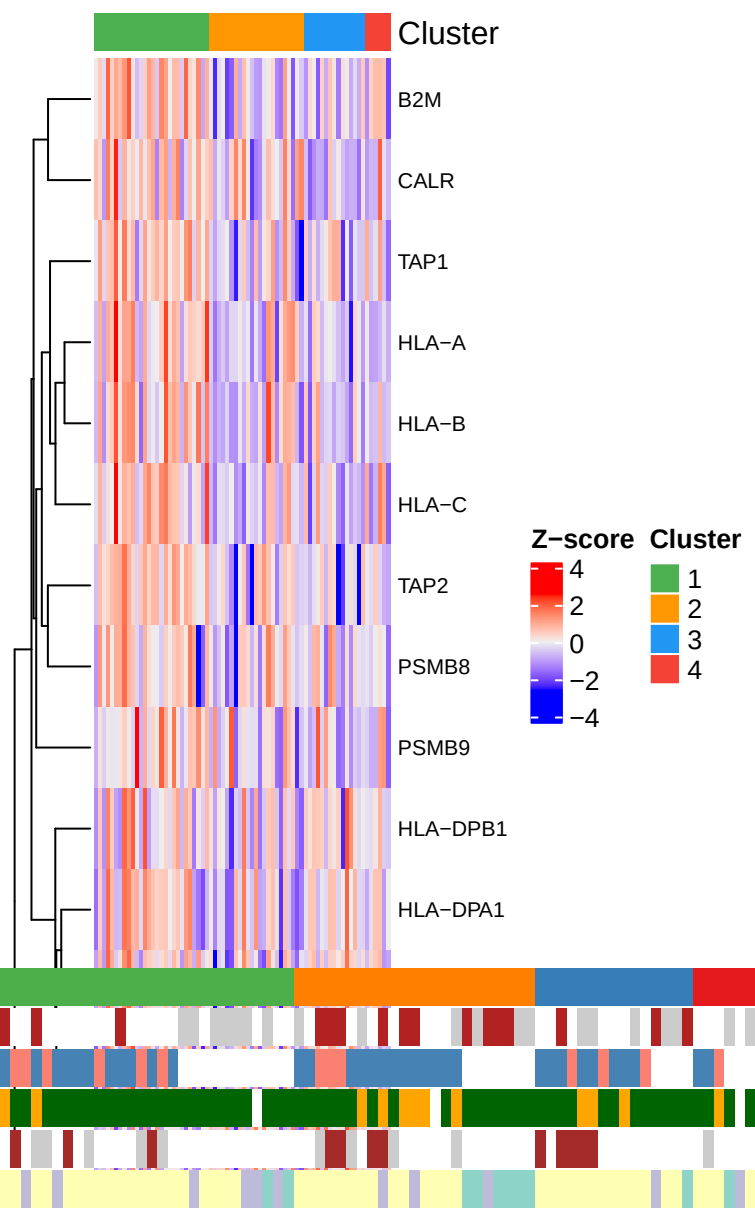
TSC2

PIK3CD

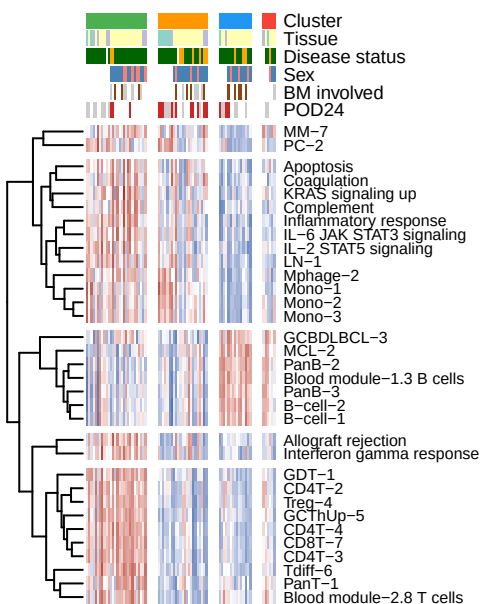
MTOR

Z-score Cluster

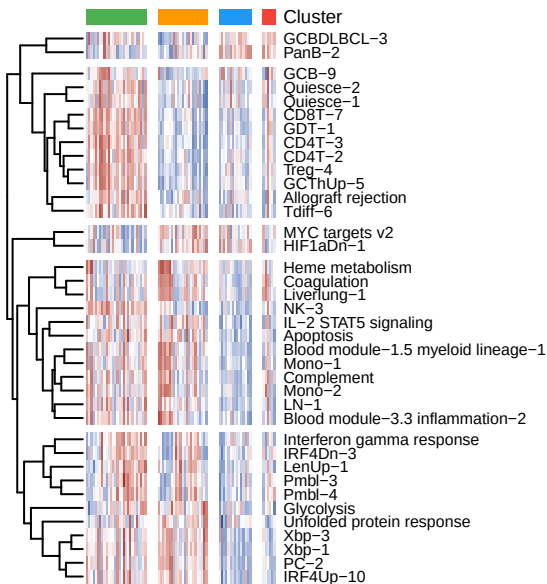




RNA



Protein



Cluster



Tissue



Disease status



Sex



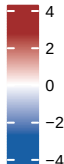
BM involved



POD24



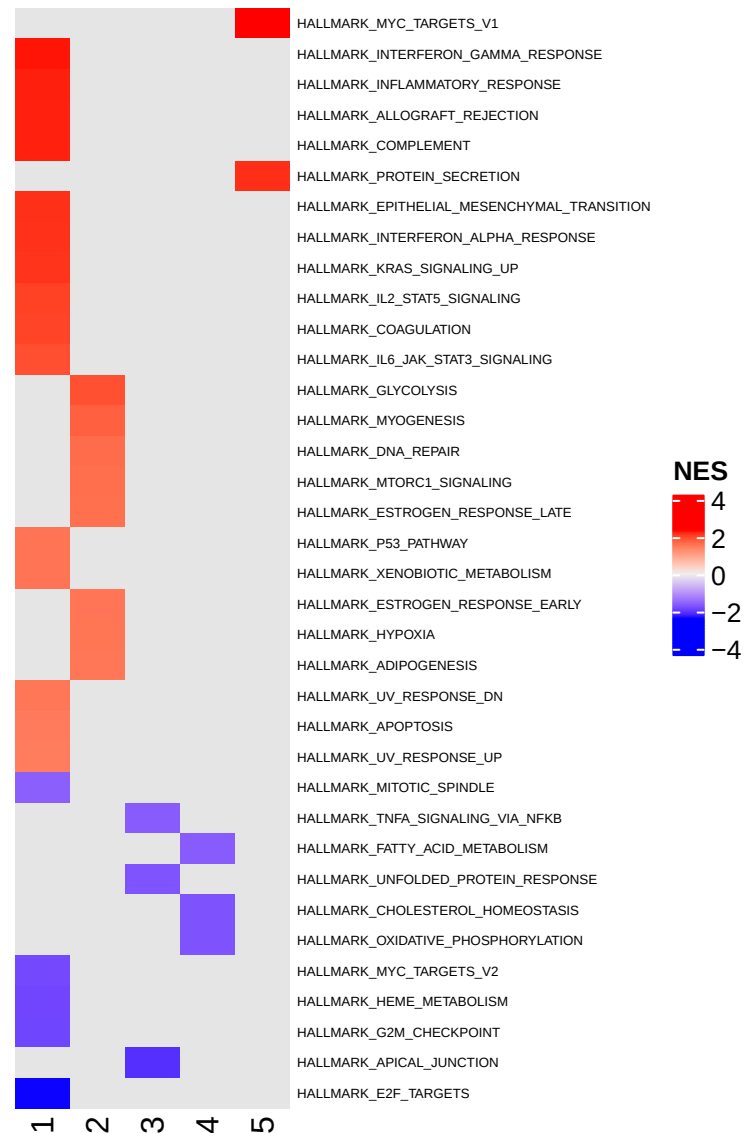
Z-score



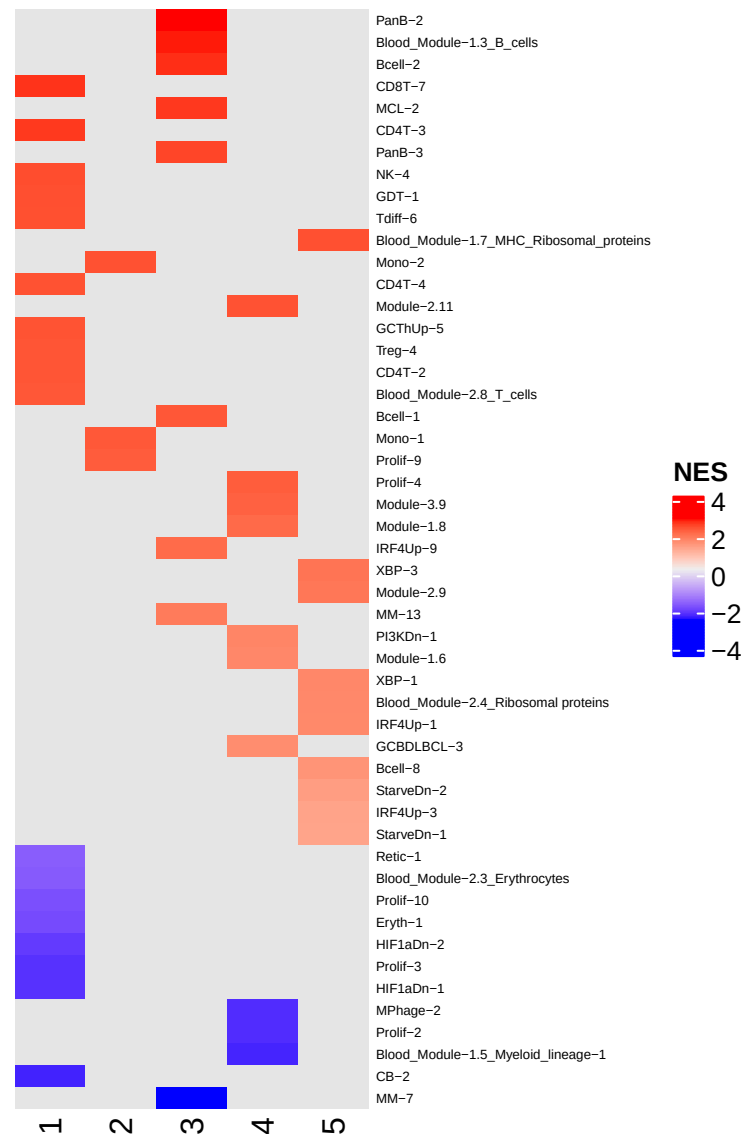
NES

4
2
0
-2
-4

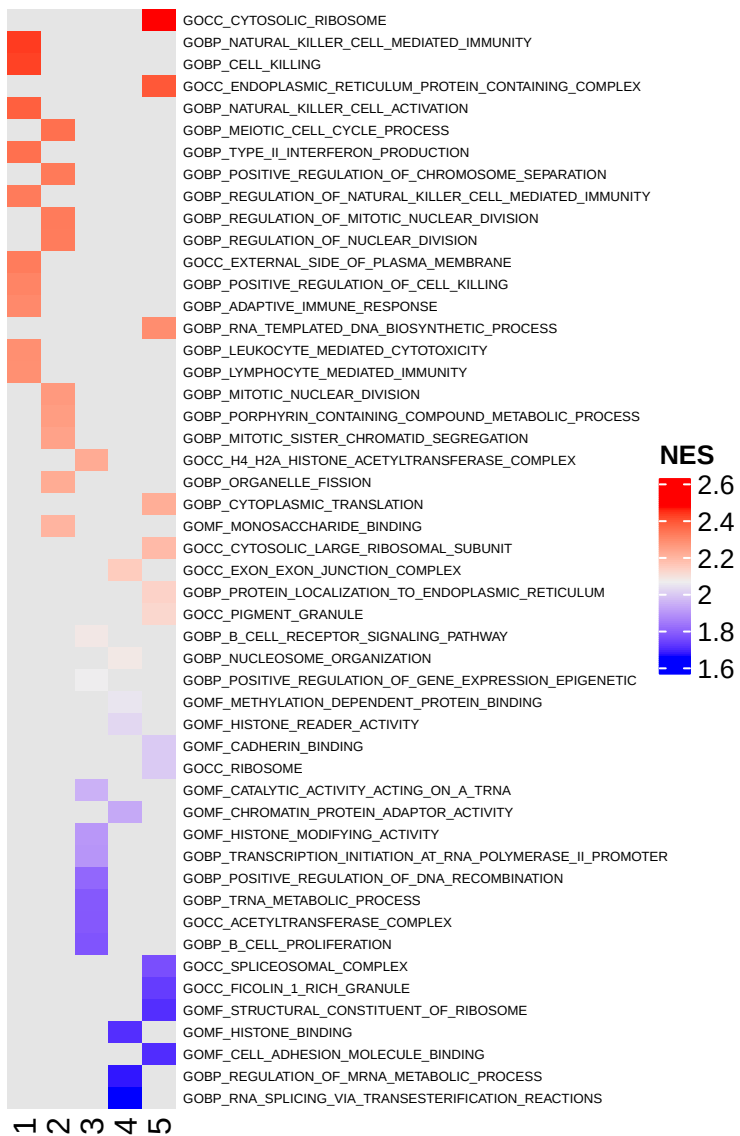
Pathway Enrichment



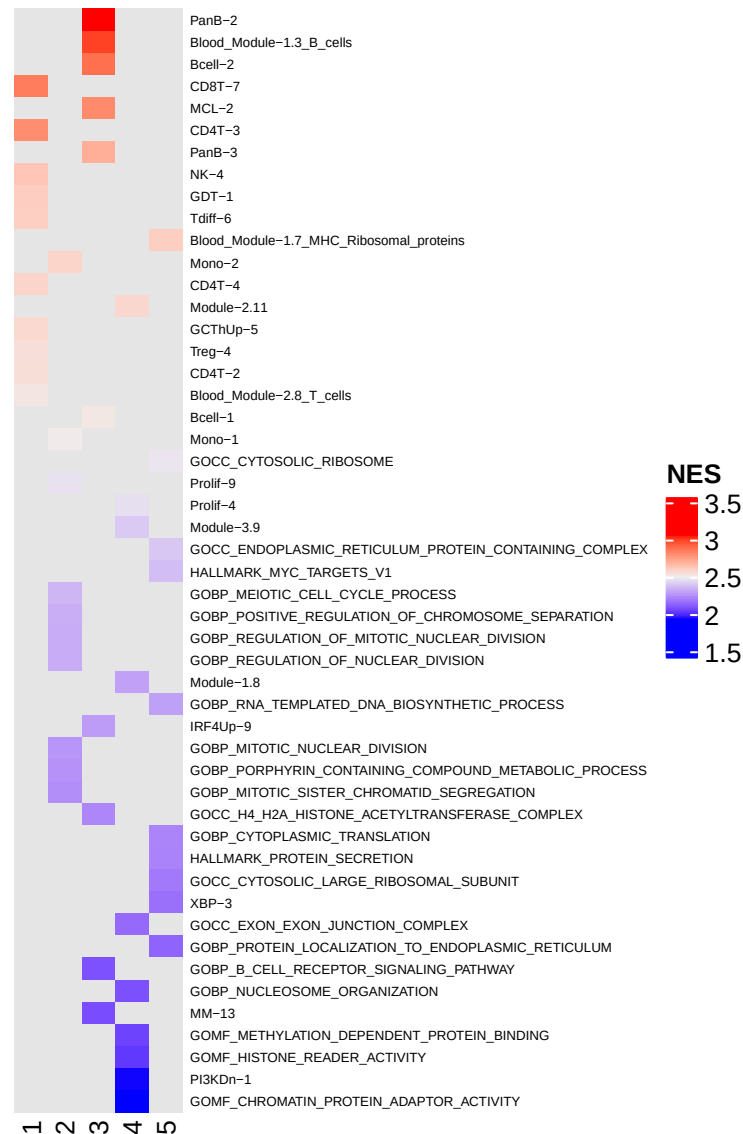
Staudt Pathways – Top 50



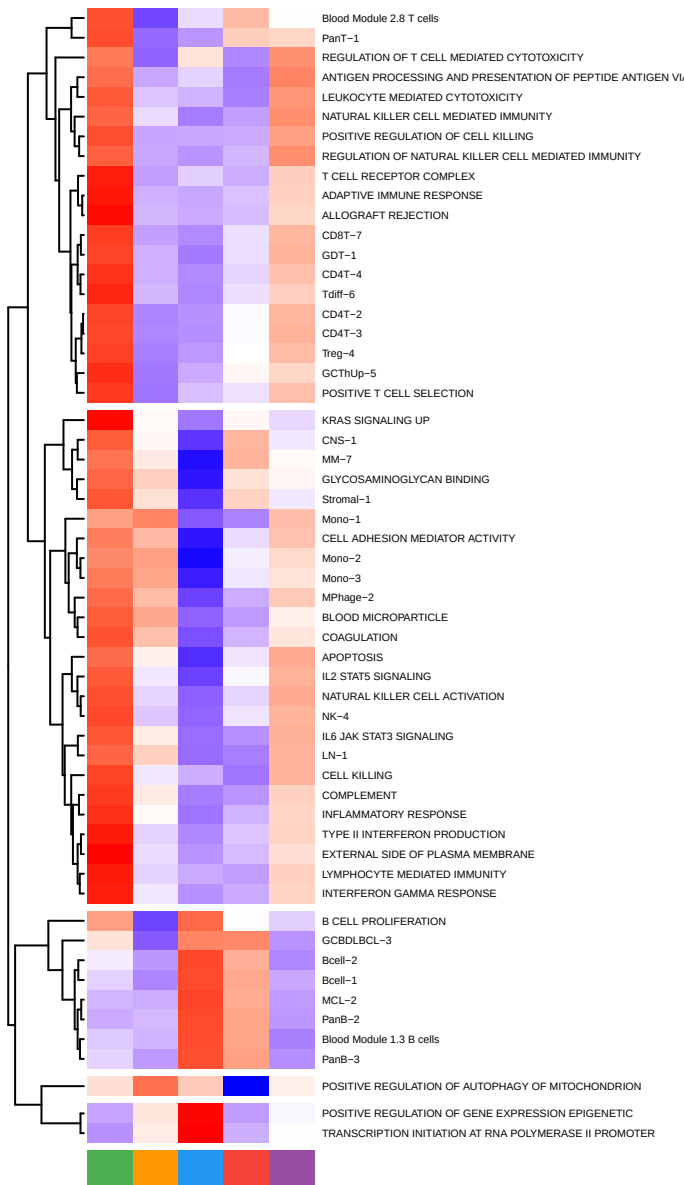
Signatures – Top 50



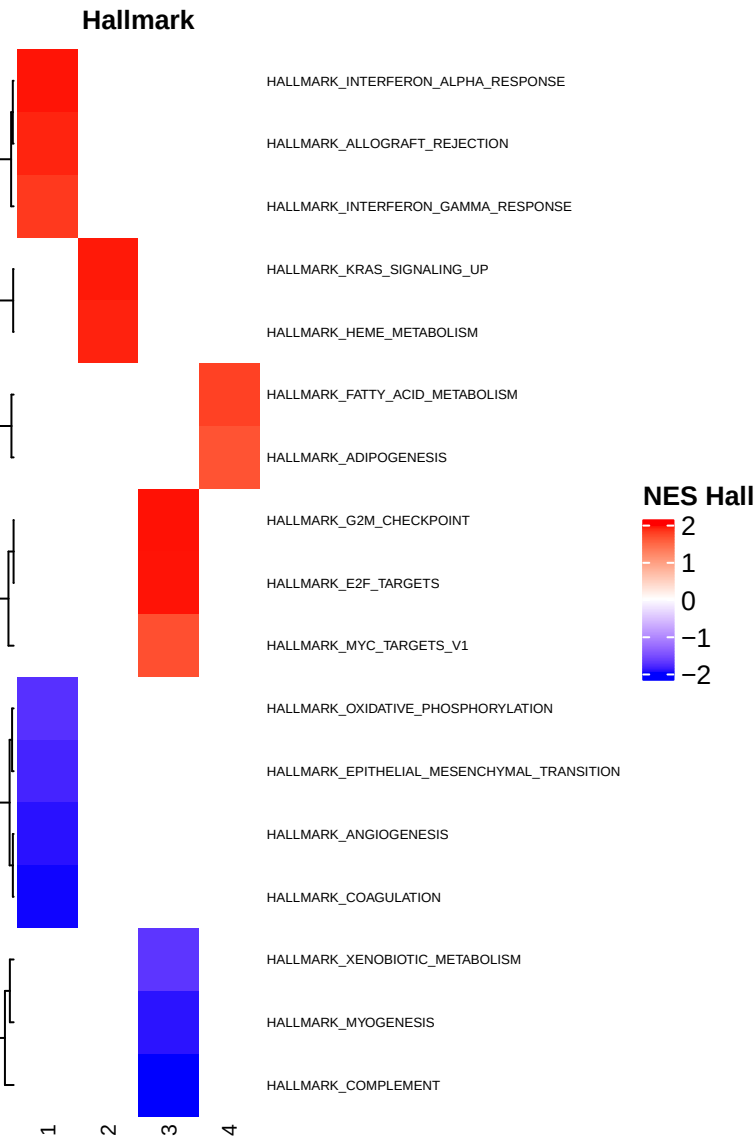
ways enriched per class



e

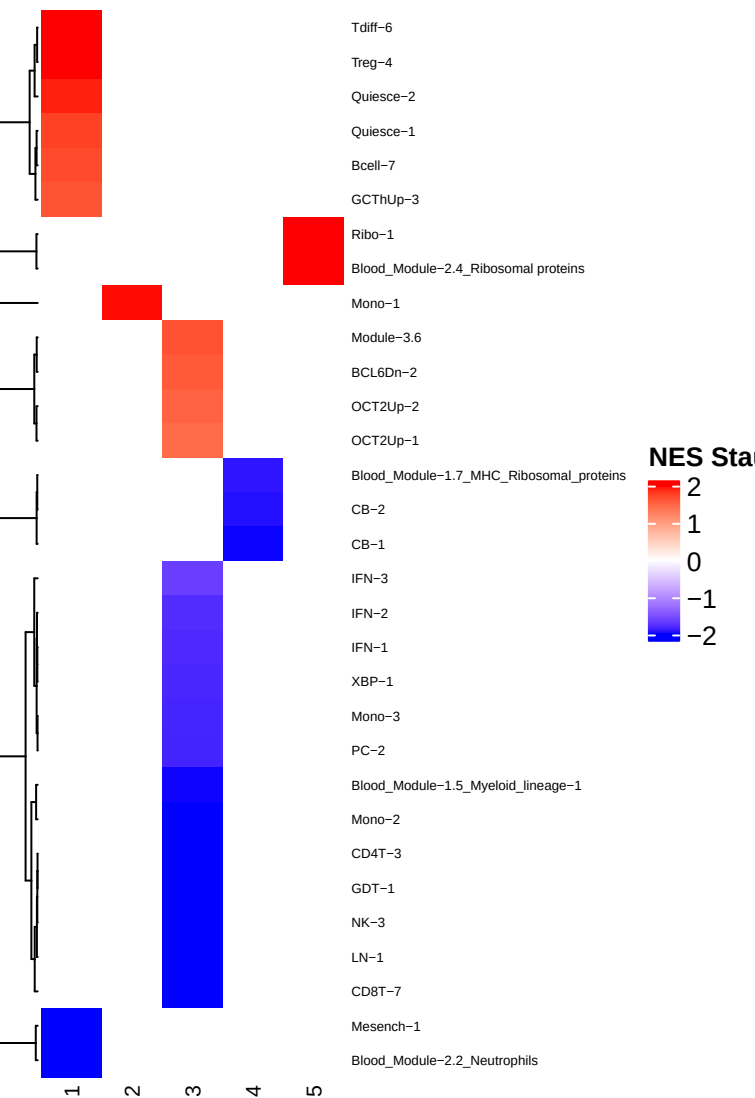


SEA Protein – Hallmark – SD:hclust k=5

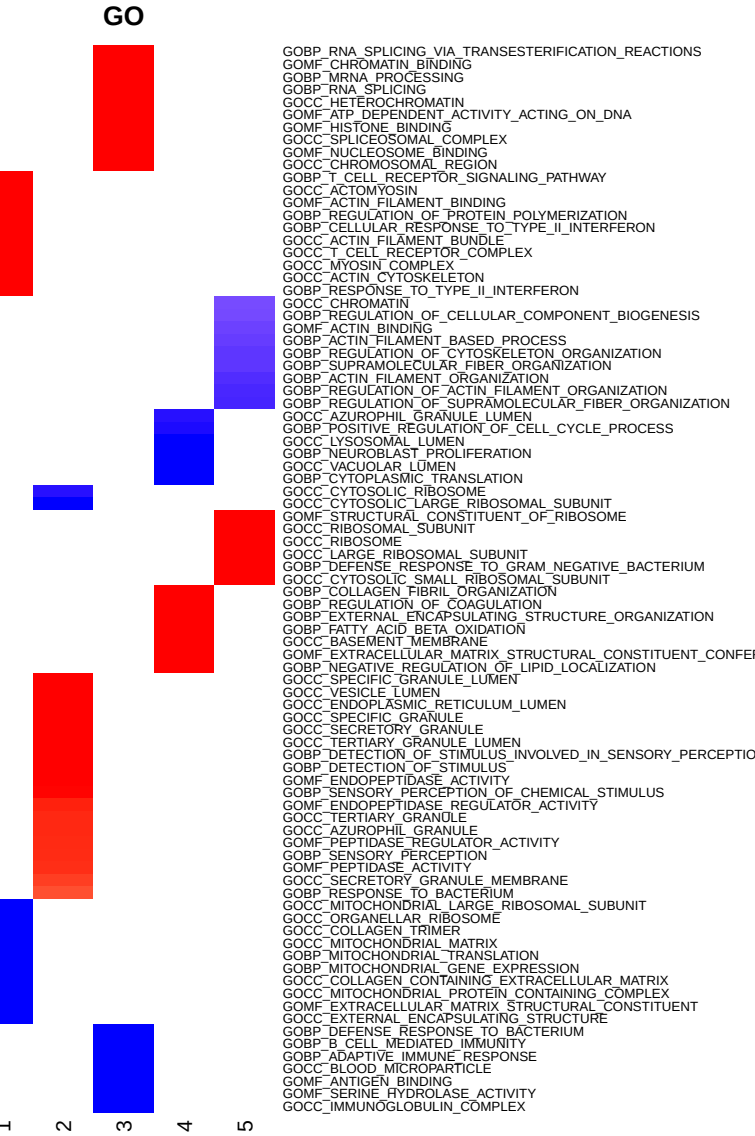


GSEA Protein – Staudt –SD:hclust k=5

Staudt



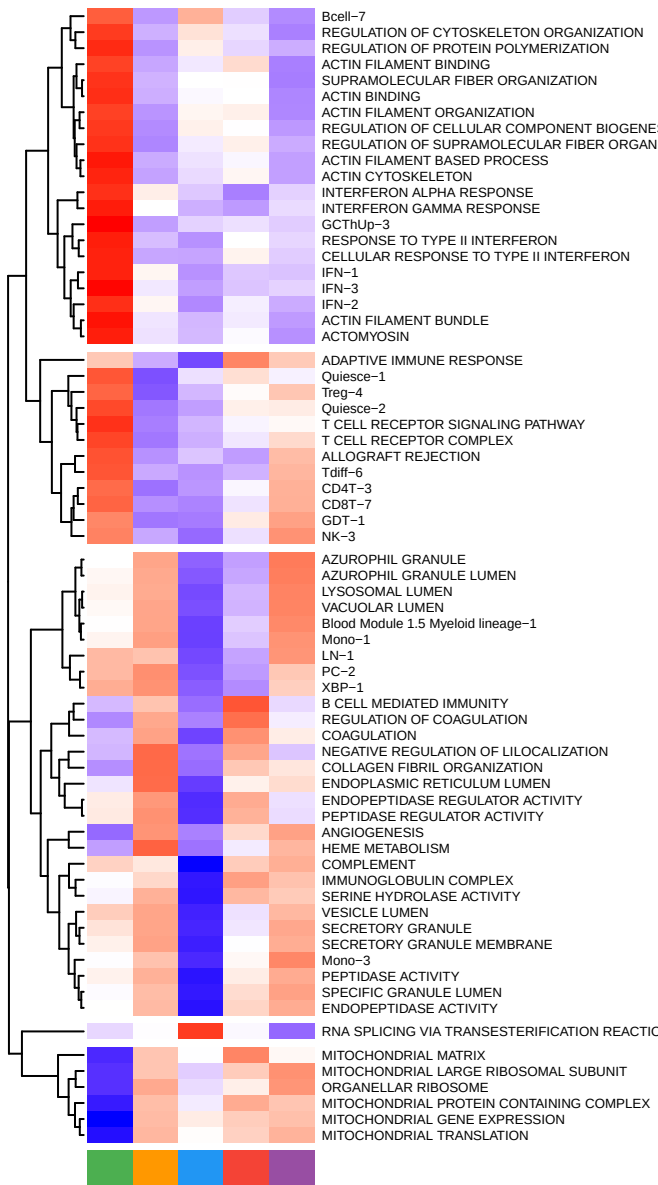
GSEA Protein – GO – SD:hclust k=5



ore

1

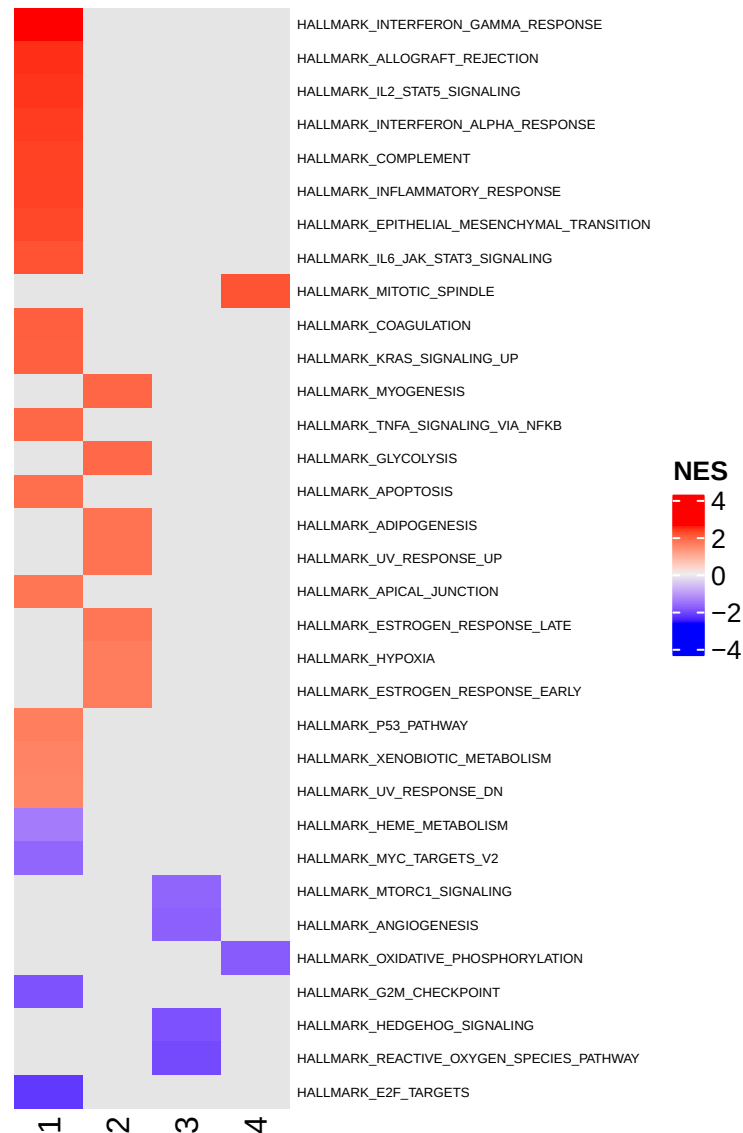
2



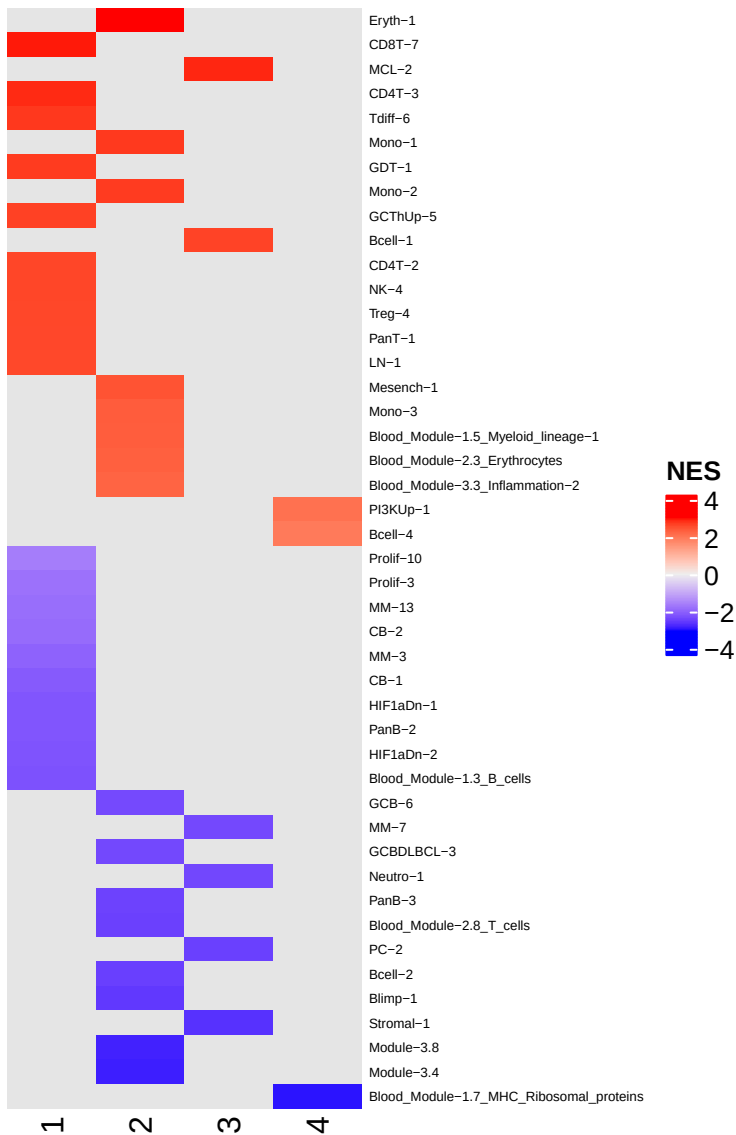
Environment Across TME Clusters



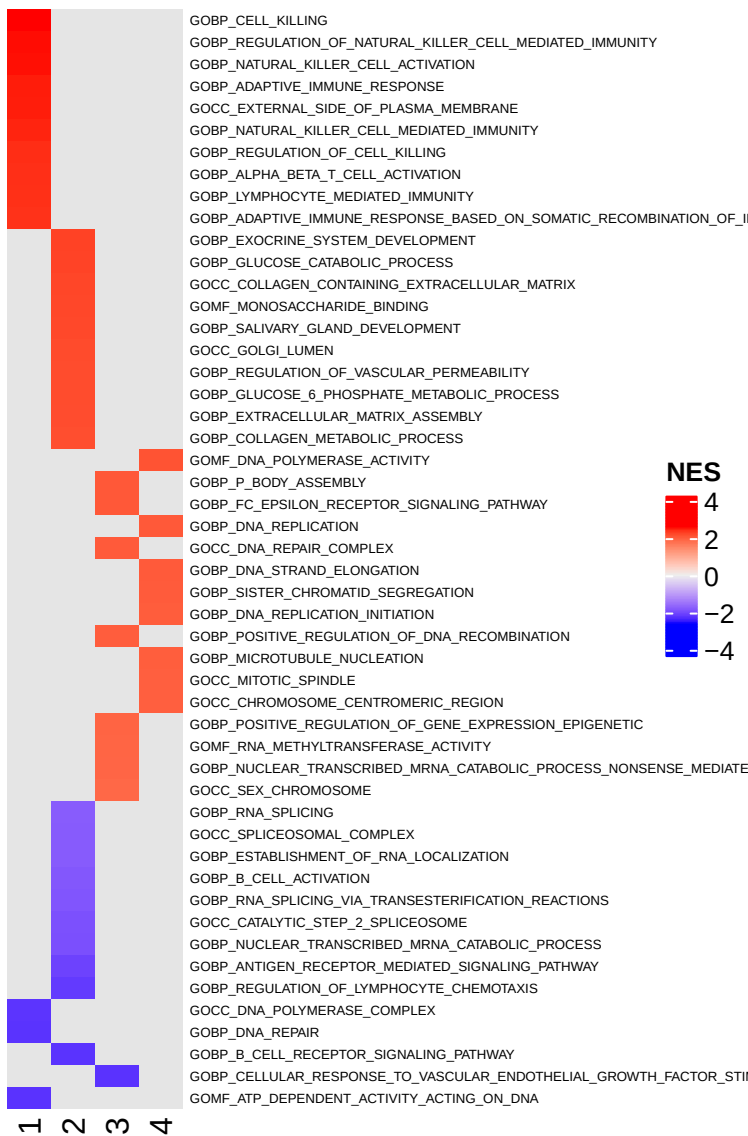
mark Pathway Enrichment



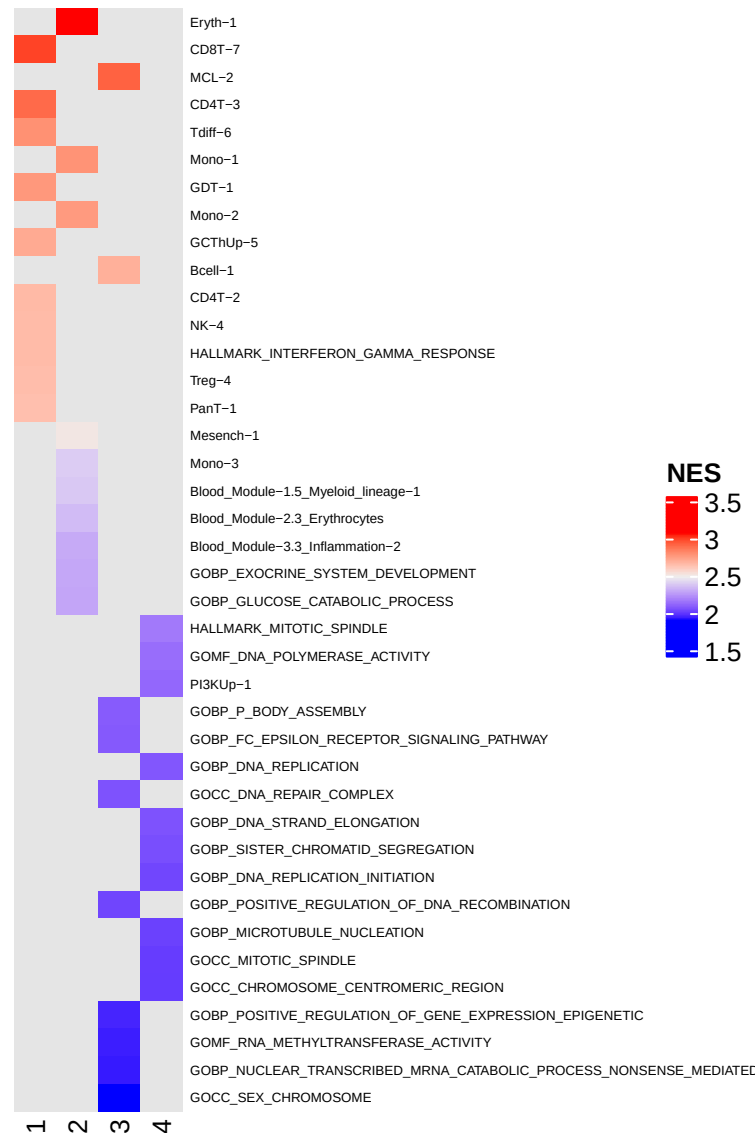
Staudt Pathways – Top 50



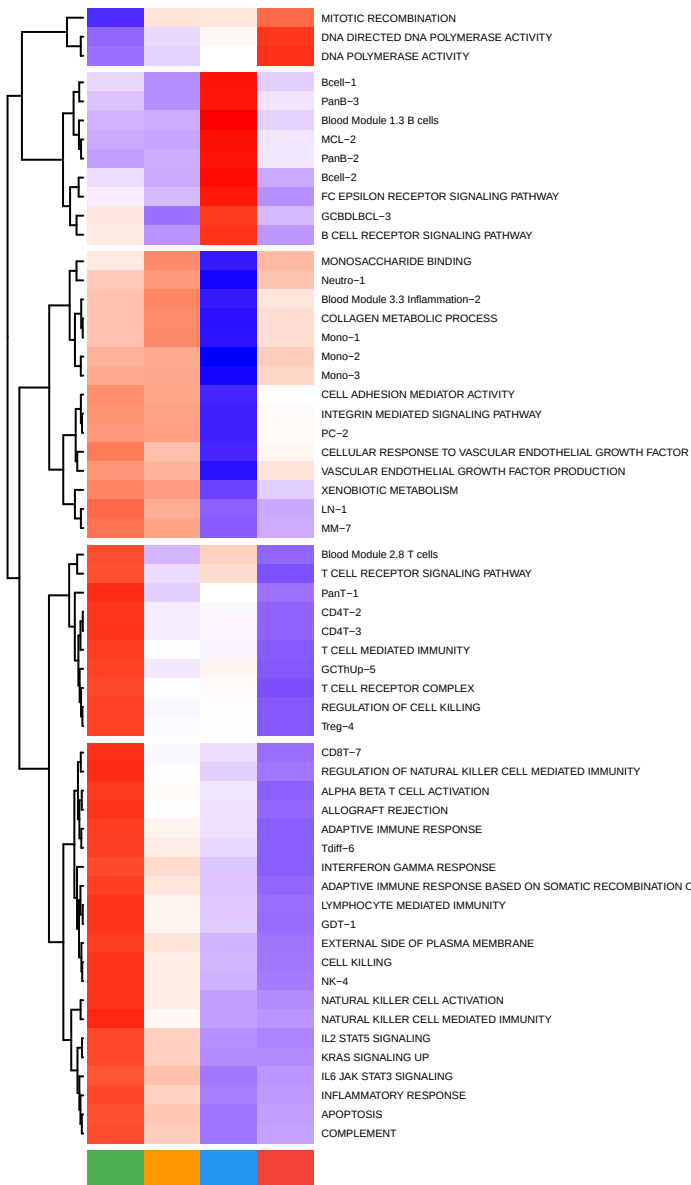
Signatures – Top 50

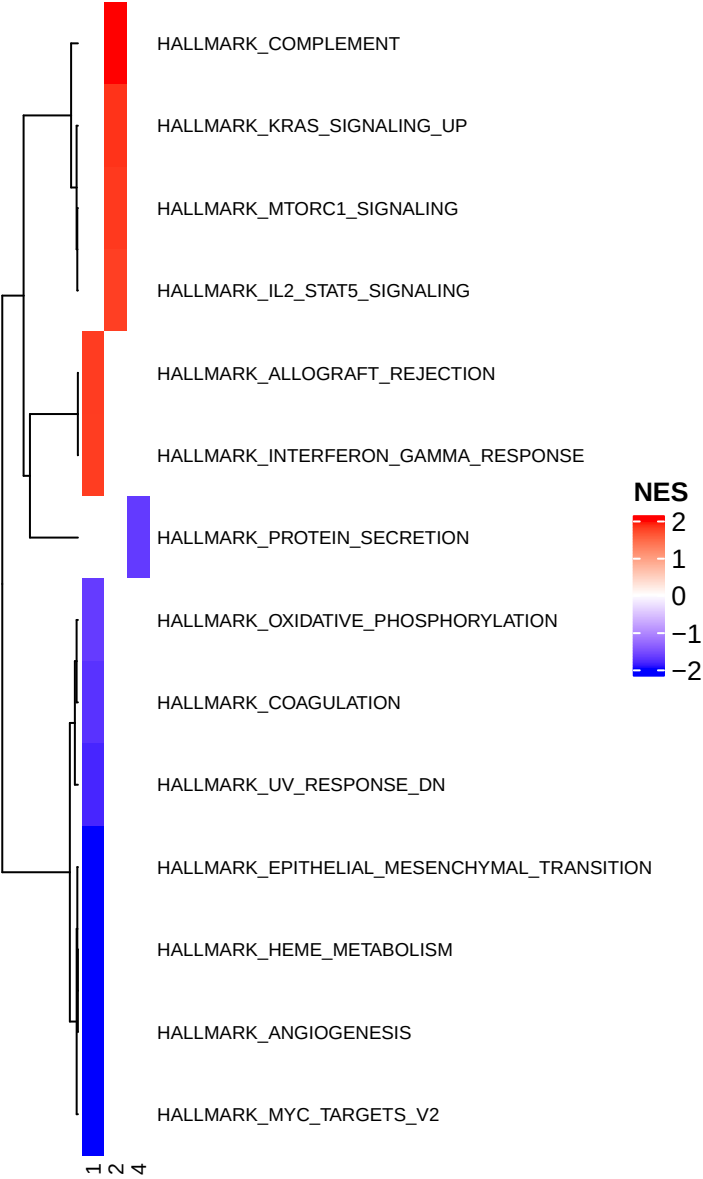


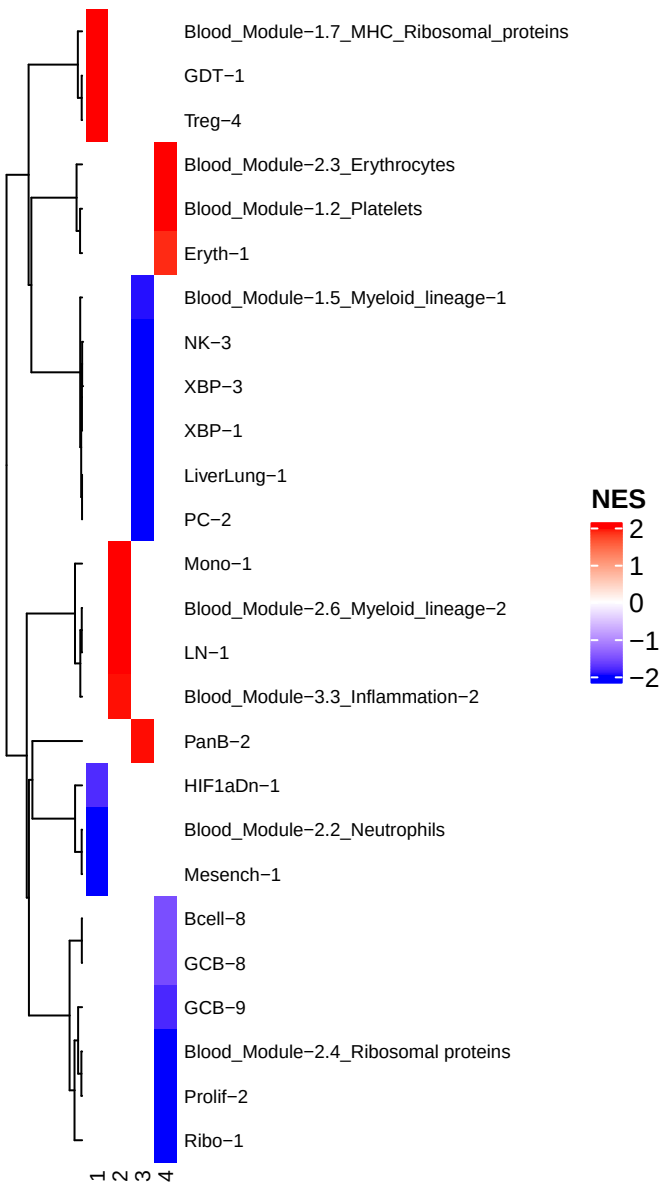
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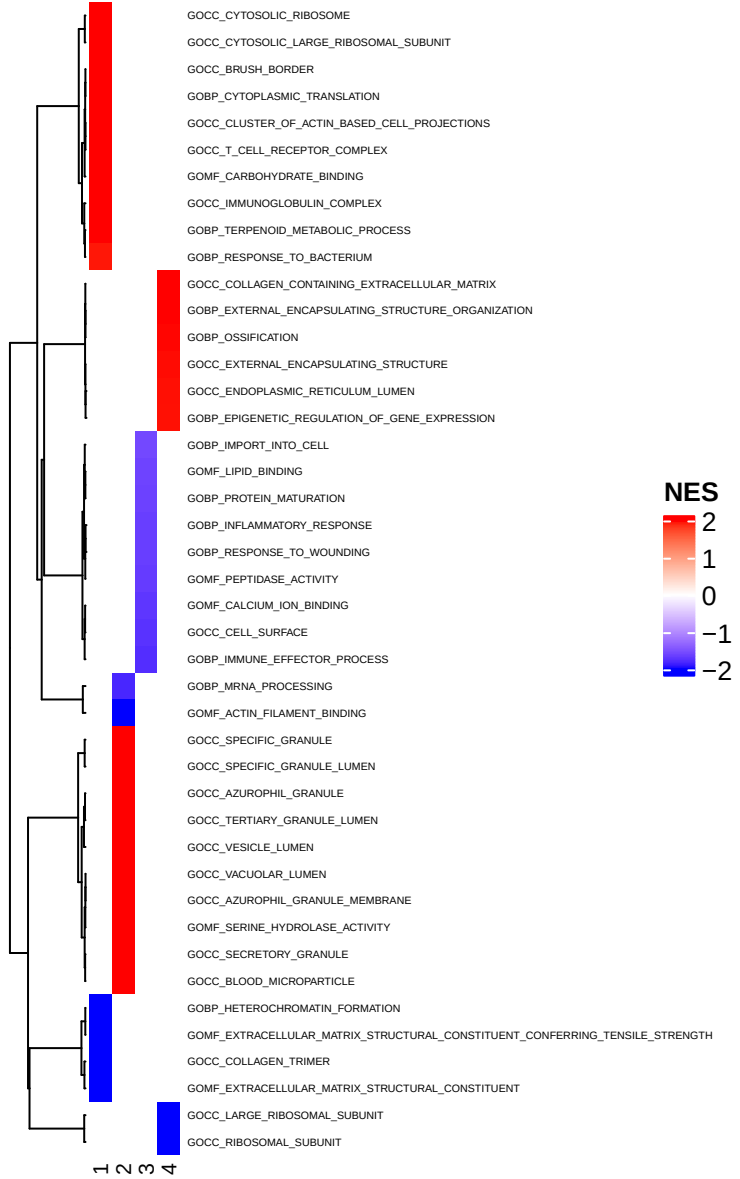


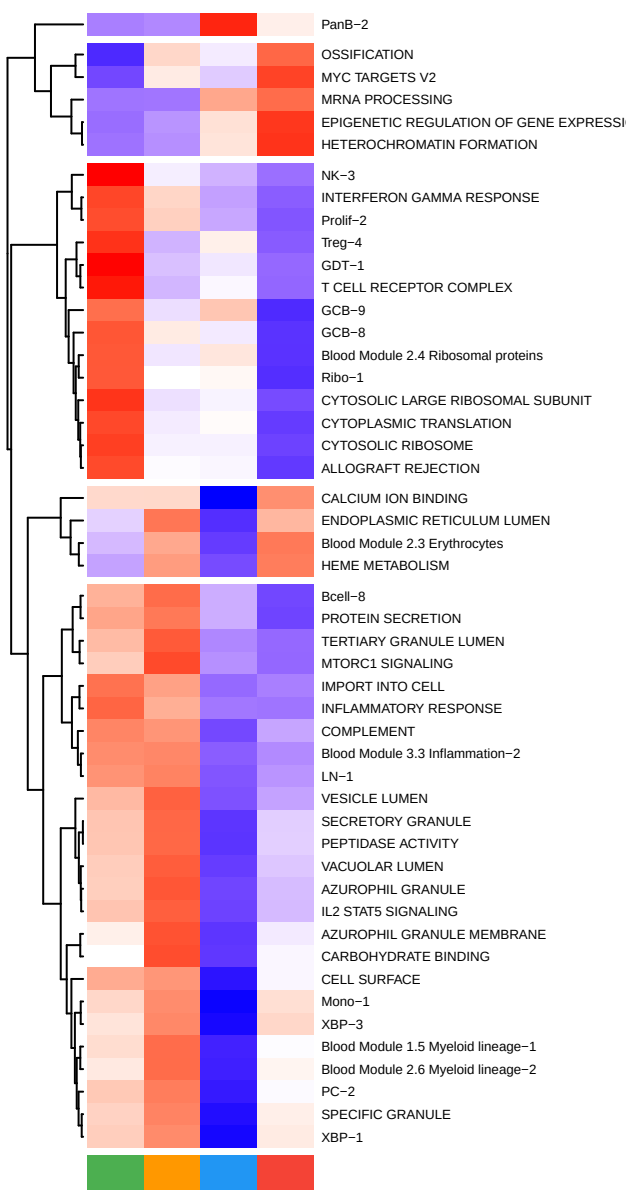
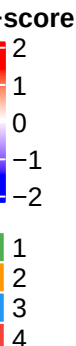
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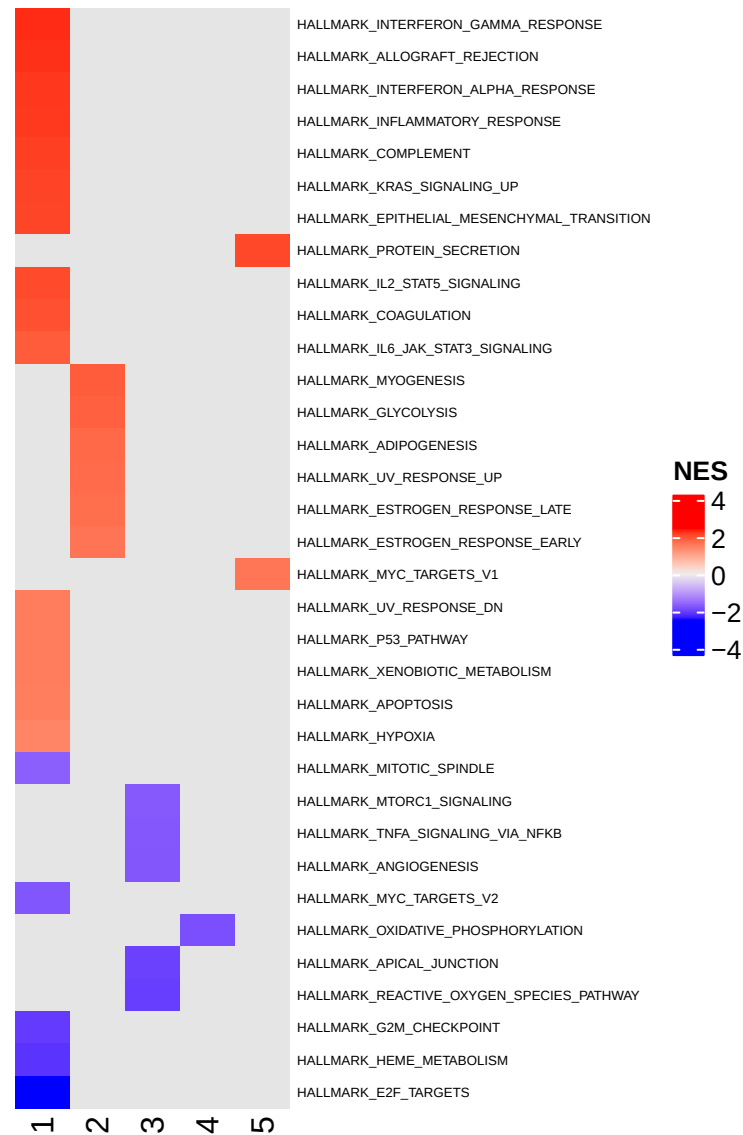
GOBP-2
GOBP-7
CD41-2
Mmp-1
Mmp-1
Gobp-1
Gobp-6
Gobp-5
Gobp-1
Tks-4
Ccl-2
Blood Module-2.8_T_cells

Neurotrophin-1
GOBP NATURAL KILLER CELL MEDIATED IMMUNITY
GOBP NATURAL_KILLER_CELL_ACTIVATION
GOBP REGULATION OF NATURAL KILLER CELL MEDIATED IMMUNITY
GOBP TUBULOSIS MEDiated IMMUNITY
GOBP ADAPTIVE IMMUNE RESPONSE
GOBP INTERFERON GAMMA RESPONSE
GOBP LYMPHOCYTE MEDIATED CYTOTOXICITY
Module-3.9
GOBP REGULATION OF CELL KILLING
GOBP -1 CELL RELEASED IMMUNITY
GOBP ALLOPATHY REJECTION
Blood Module-2.8_Humoral Immunity
GOBP EXOCRINE SYSTEM DEVELOPMENT
GOBP MONOSACCHARIDE BINDING
GOBP GLUCOSE TRANSPORT PHOSPHATE_BINDING
GOBP GLUCOSE CATABOLIC PROCESS
GOBP PROMOTER GRANULE
GOBP INTERFERON ALPHA RESPONSE
GOBP COLLAGEN CONTAINING EXTRACELLULAR MATRIX
GOBP ADAPTIVE PROTEIN FOLDING_CHAPERONE
GOBP ADAPTIVE BINDING
GOBP COLLAGEN METABOLIC PROCESS
GOBP COMPLEMENT
GOBP REGULATION OF PHOSPHATE METABOLIC PROCESS
GOBP REGULATOR OF VASCULAR PERMEABILITY
GOBP DNA POLYMERASE ACTIVITY
GOBP G-protein coupled receptor signaling pathway
GOBP PC-1000 RECEPTOR SIGNALING PATHWAY
GOBP EPITHELIAL MESENCHYMAL TRANSITION
GOBP METABOLISM OF ENDOPLASMIC RETICULUM
GOBP PROTEIN SORTITION
GOBP DE NOVO POST TRANSLATIONAL PROTEIN FOLDING
GOBP PROTEIN FOLDING CHAPERONE
GOBP COLLAGEN RICH GRANULE
GOBP DNA STRAND SEQUENCING
Kip-1
GOBP DNA REPLICATION
GOBP COAGULATION
GOBP ESTROGEN SECRETION SEGREGATION
GOBP DNA DIRECTED DNA POLYMERASE ACTIVITY
GOBP DNA REPLICATION
GOBP MITOCHONDRIAL NUCLEATION
GOBP MITOCHONDRIAL DNA RECOMBINATION
GOBP CHROMOSOME CENTROMERIC REGION
GOBP NUCLEAR TRANSCRIBED MRNA CATABOLIC_PROCESS_NONSENSE_MEDIATED_DECAY
GOBP NUCLEAR TUBES AND SIGNALING
GOBP POSITIVE REGULATION OF GENE EXPRESSION_EPIGENETIC
GOBP METABOLIC TRANSFERENCE ACTIVITY
GOBP INDEPENDENT ACTIVITY_ACTING_ON_DNA
GOBP ADIPOGENESIS
GOBP
GOBP UV RESPONSE UP
GOBP ESTROGEN RESPONSE LATE
GOBP SECRETORY GRANULE MEMBRANE
GOBP ANTIGEN PRESENTATION RESPONSE EARLY
GOBP TARGETS_V1
GOBP COATED VESICLE
GOBP BILAYER VESICLE
GOBP UV RESPONSE DN
GOBP EXOTOSIS
GOBP TGF-beta PATHWAY
GOBP CELL SURVIVAL-METABOLISM
GOBP APOPTOSIS
GOBP CELL SUBSTRATE JUNCTION
GOBP PROTEIN MATURATION
GOBP CELL MEMBRANE
GOBP HYPOXIA
Nrf-1
GOBP MITOTIC SPINDLE
GOBP-1
GOBP MTORC1_SIGNALING
GOBP DNA BINDING TRANSCRIPTION ACTIVATOR_ACTIVITY
GOBP TNFA SIGNALING VIA NFkB
GOBP ANGIOGENESIS
GOBP MYC_TARGETS_V2
GOBP ESTABLISHMENT OF RNA LOCALIZATION
GOBP SPLICEDOMAIN COMPLEX
GOBP OXIDATIVE PHOSPHORYLATION
GOBP TRANSCRIPTION OF MRNA METABOLIC_PROCESS
GOBP CELL ACTIVATION
GOBP RNA SPLICING VIA TRANSFERIFICATION_REACTIONS
Blood Module-2.8_Lymphocytes
GOBP MITOCHONDRIAL PROTEIN BIOSYNTHESIS
GOBP NUCLEAR TRANSCRIBED MRNA CATABOLIC_PROCESS
MM-15
MM-2
GOBP APICAL JUNCTION
GOBP REACTIVE OXYGEN SPECIES PATHWAY
GOBP GSK3 CHECKPOINT
GOBP ANTIGEN PRESENTATION MEDIATED SIGNALING_PATHWAY
GOBP HEME METABOLISM
GOBP-1
GOBP REGULATION OF LYMPHOCTE_CHEMOTAXIS
GOBP-3
GOBP PROTEOGLYCAN BINDING
GOBP REGULATION OF CHROMOSOME SEGREGATION
GOBP UNREPAIRED DNA REPLICATION_MAINTENANCE_OF_FIDELITY
GOBP GROWTH FACTOR BINDING
GOBP CELLULAR RESPONSE TO VASCULAR ENDOTHELIAL_GROWTH_FACTOR_STIMULUS
GOBP DNA TOPOLATASE II
GOBP DNA REPAIR
GOBP REGULATION OF SISTER CHROMATID SEGREGATION
GOBP CELLULAR ENDOTHELIAL_GROWTH_FACTOR_PRODUCTION
GOBP T CELL ANTIGEN PRESENTATION SIGNALING PATHWAY
GOBP INTEGRIN MEDIATED SIGNALING PATHWAY
GOBP ANGIOGENESIS
GOBP REGULATION OF MITOTIC_SISTER_CHROMATID_SEGREGATION
GOBP MITOTIC_ACTIVITY_ACTING_ON_DNA
Neurotrophin-1
GOBP-3
Blood Module-1.3_B_cells

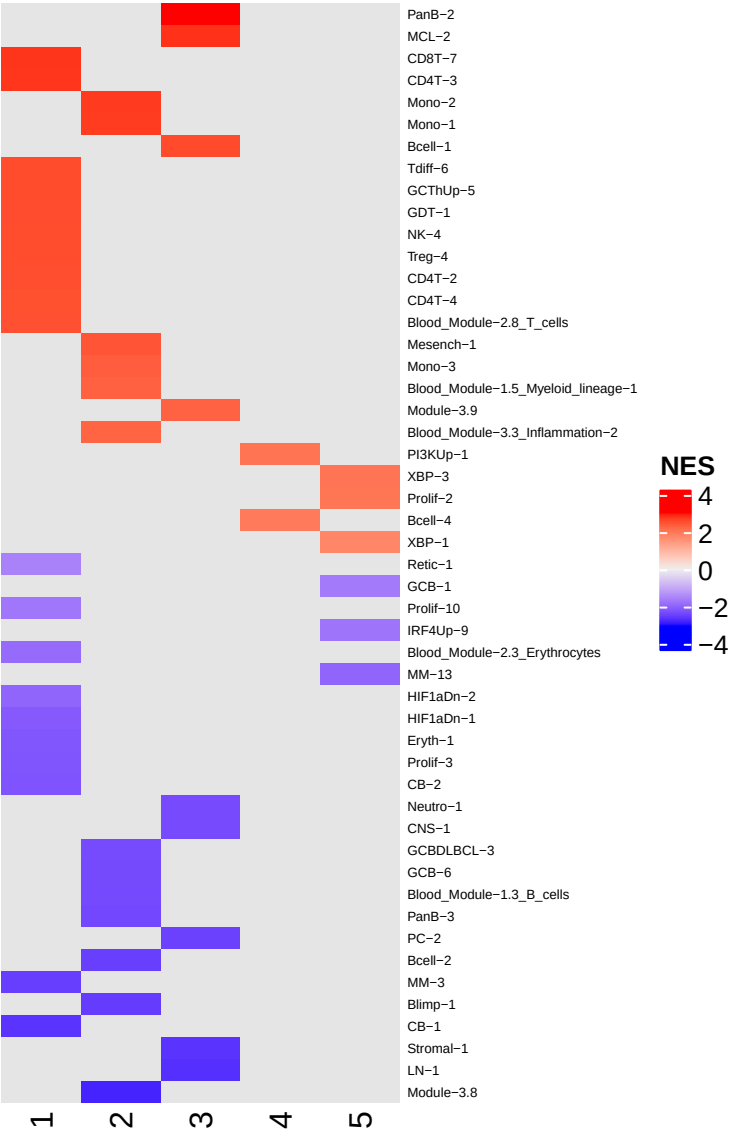
GOBP REGULATION OF ATTACHMENT_OF_SPINDLE_MICROTUBULES_TO_KINETOCHORE
GOBP E2F_TARGETS
GOBP-2
GOBP EXTERNAL_ENCAPSULATING_STRUCTURE
MM-2
GOBP ATTACHMENT OF SPINDLE MICROTUBULES TO KINETOCHORE
GOBP ANTIGEN PRESENTATION OF EXOGENOUS PEPTIDE_ANTIGEN
GOBP T CELL RECEPTOR SIGNALING PATHWAY
GOBP ANTIGEN PROCESSING AND PRESENTATION_OF_EXOGENOUS_ANTIGEN
GOBP CYTOSOLIC_LARGE_RIBOSOMAL_SUBUNIT
GOBP
GOBP CYTOPLASMIC TRANSLATION
GOBP ANTIGEN PROCESSING AND PRESENTATION_OF_PEPTIDE_ANTIGEN
Module-3.8
Module-3.4
GOBP CYTOSOLIC RIBOSOME
Blood Module-1.3_Pituitary proteins

1 2 3 4 5

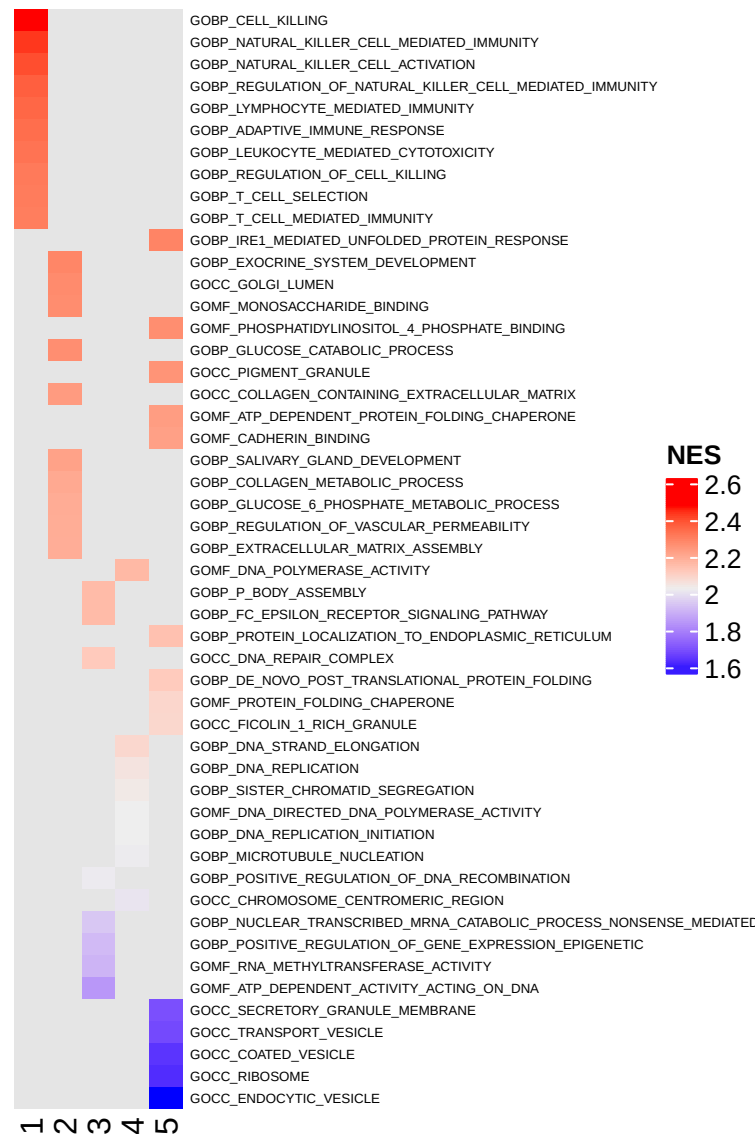
mark Pathway Enrichment



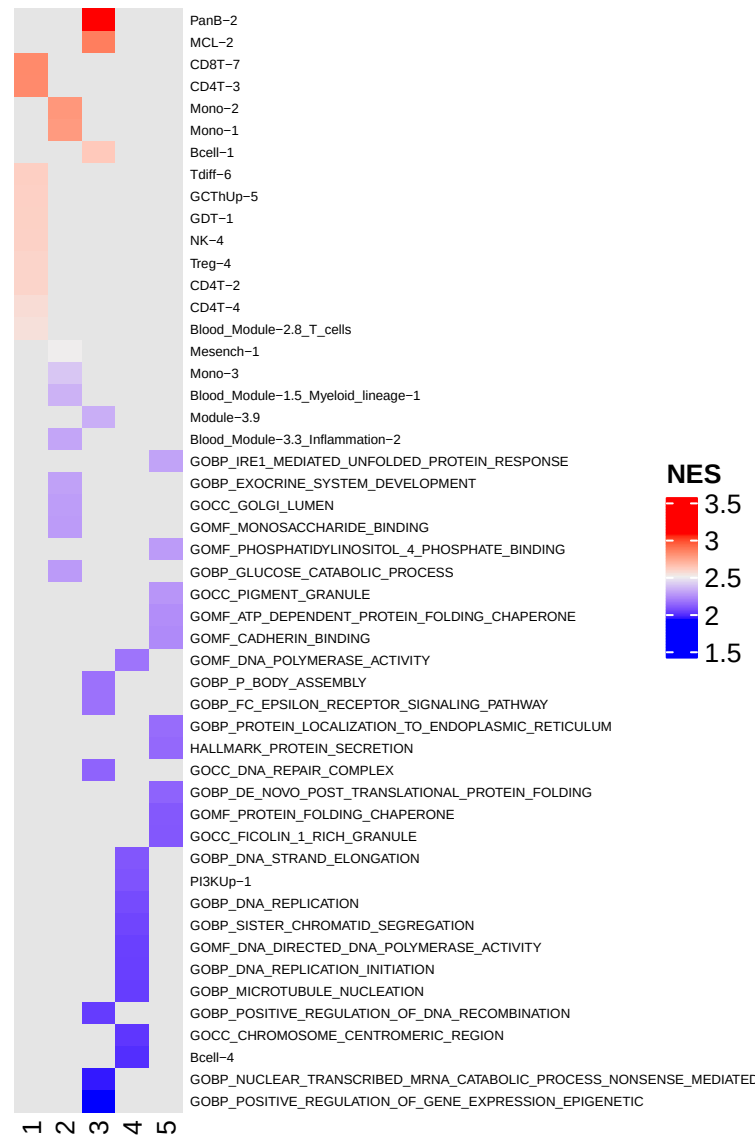
Staudt Pathways – Top 50



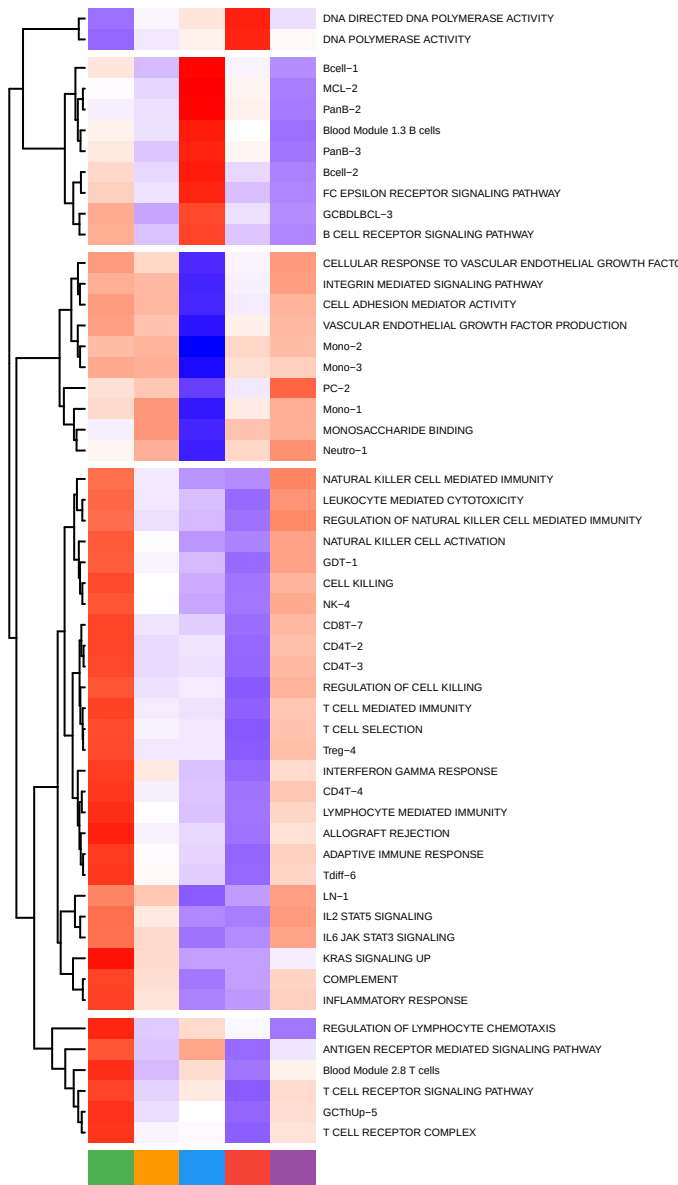
Signatures – Top 50



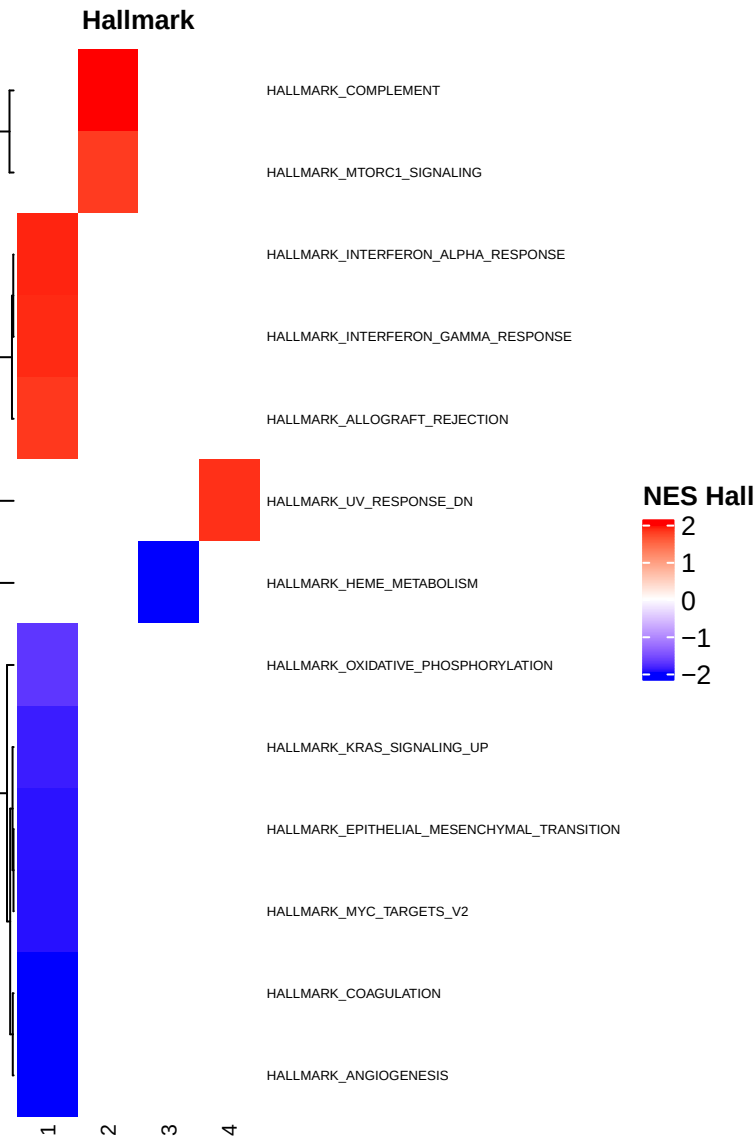
ways enriched per class



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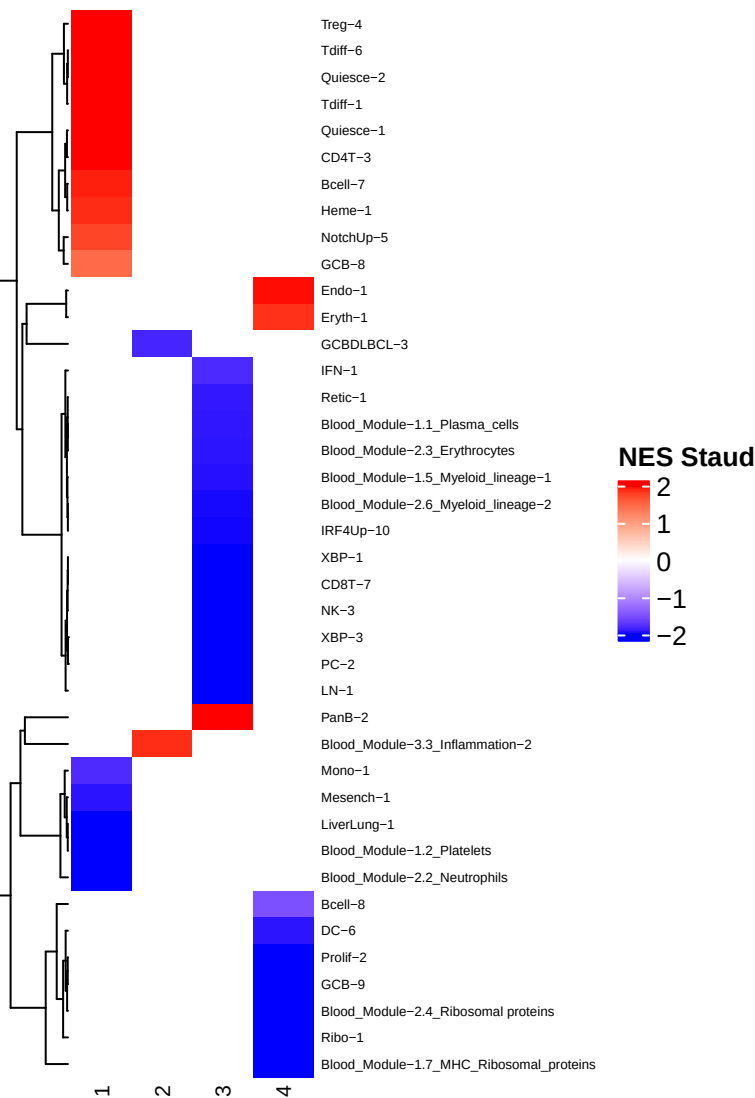


SEA Protein – Hallmark – SD:hclust k=5



GSEA Protein – Staudt –SD:hclust k=5

Staudt



GSEA Protein – GO – SD:hclust k=5

GO



